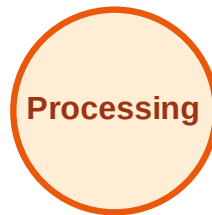


BFS Traversal

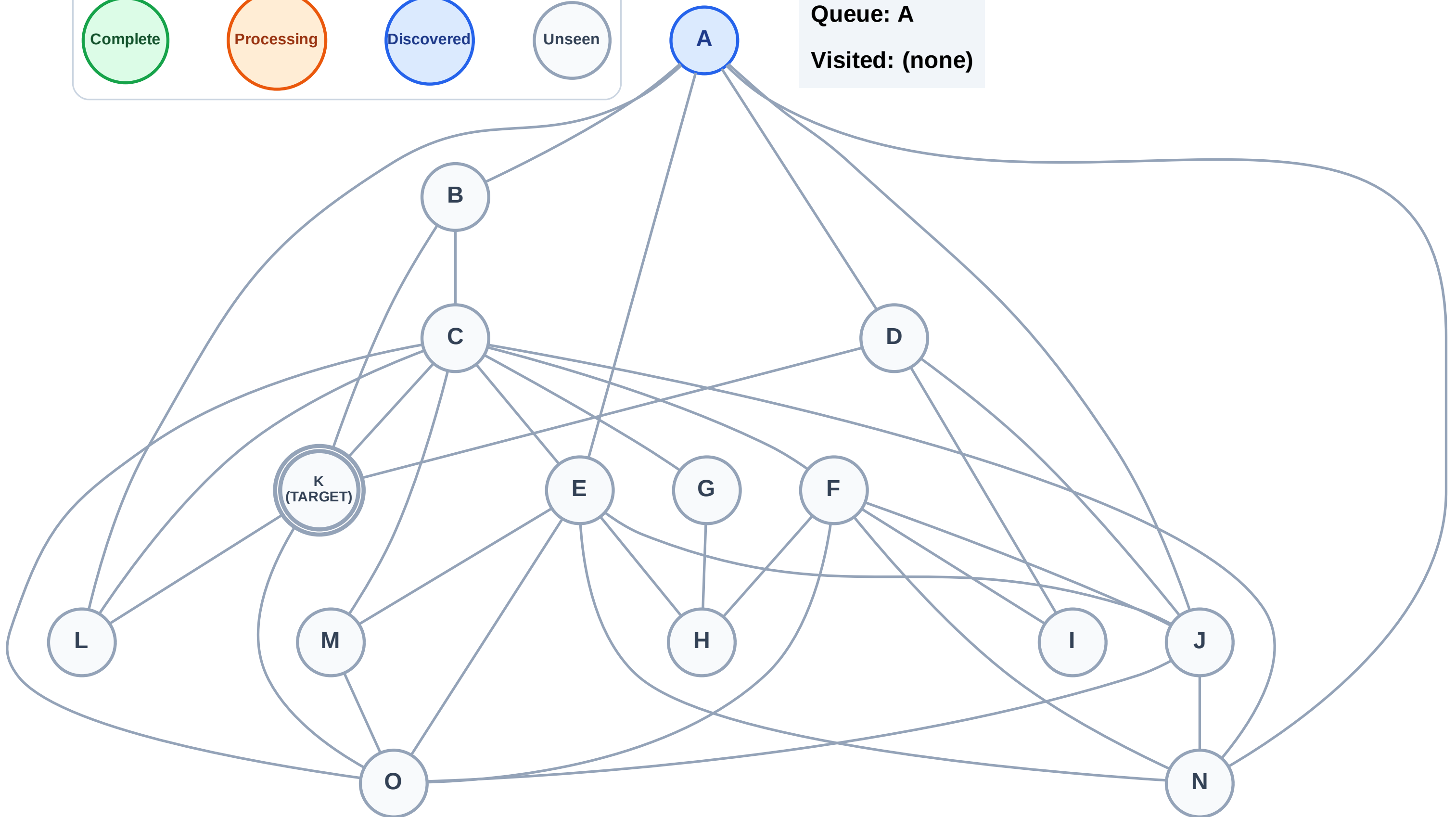
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

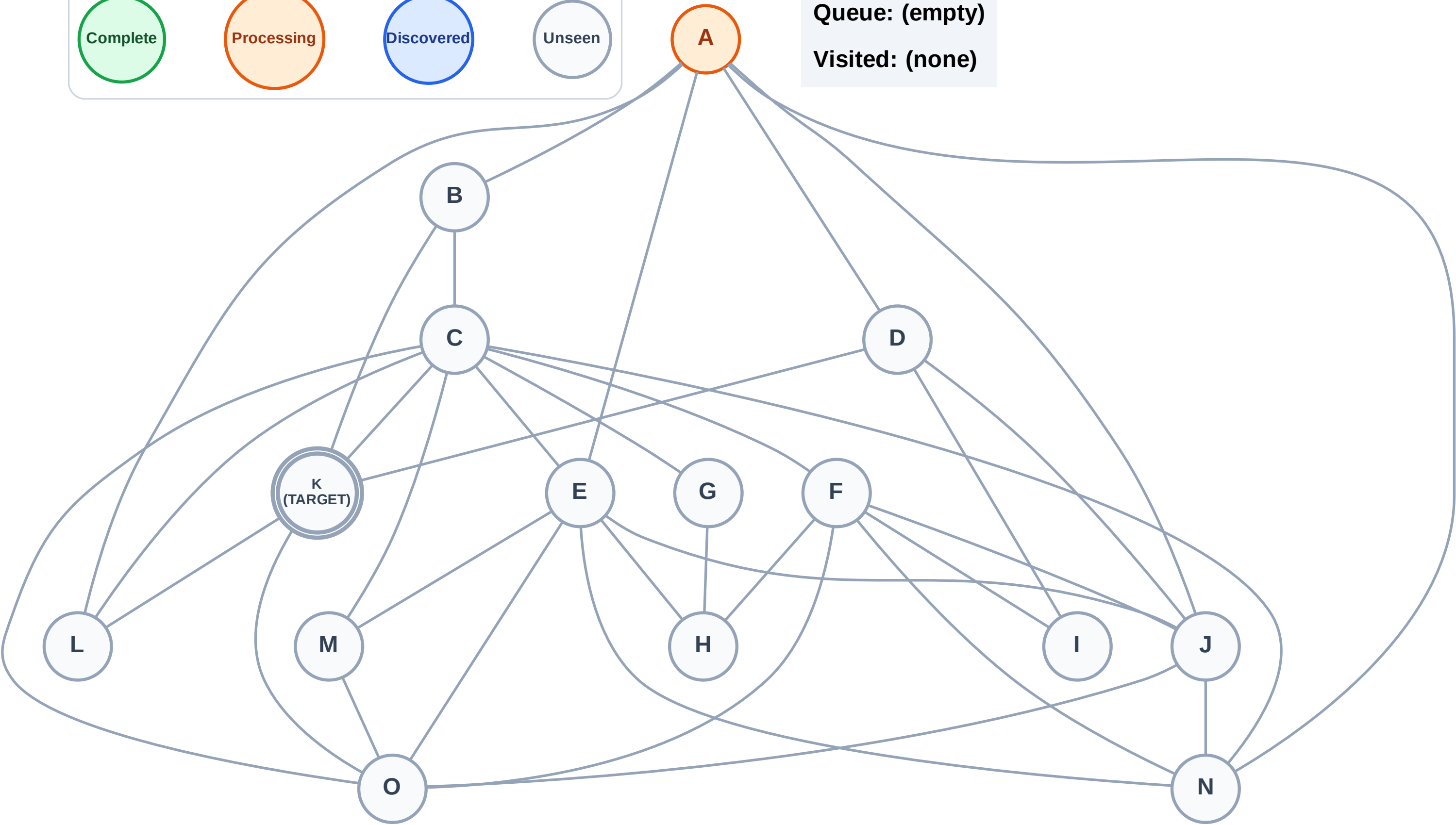
Complete

Processing

Discovered

Unseen

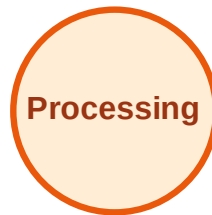
Queue: (empty)
Visited: (none)



BFS Traversal

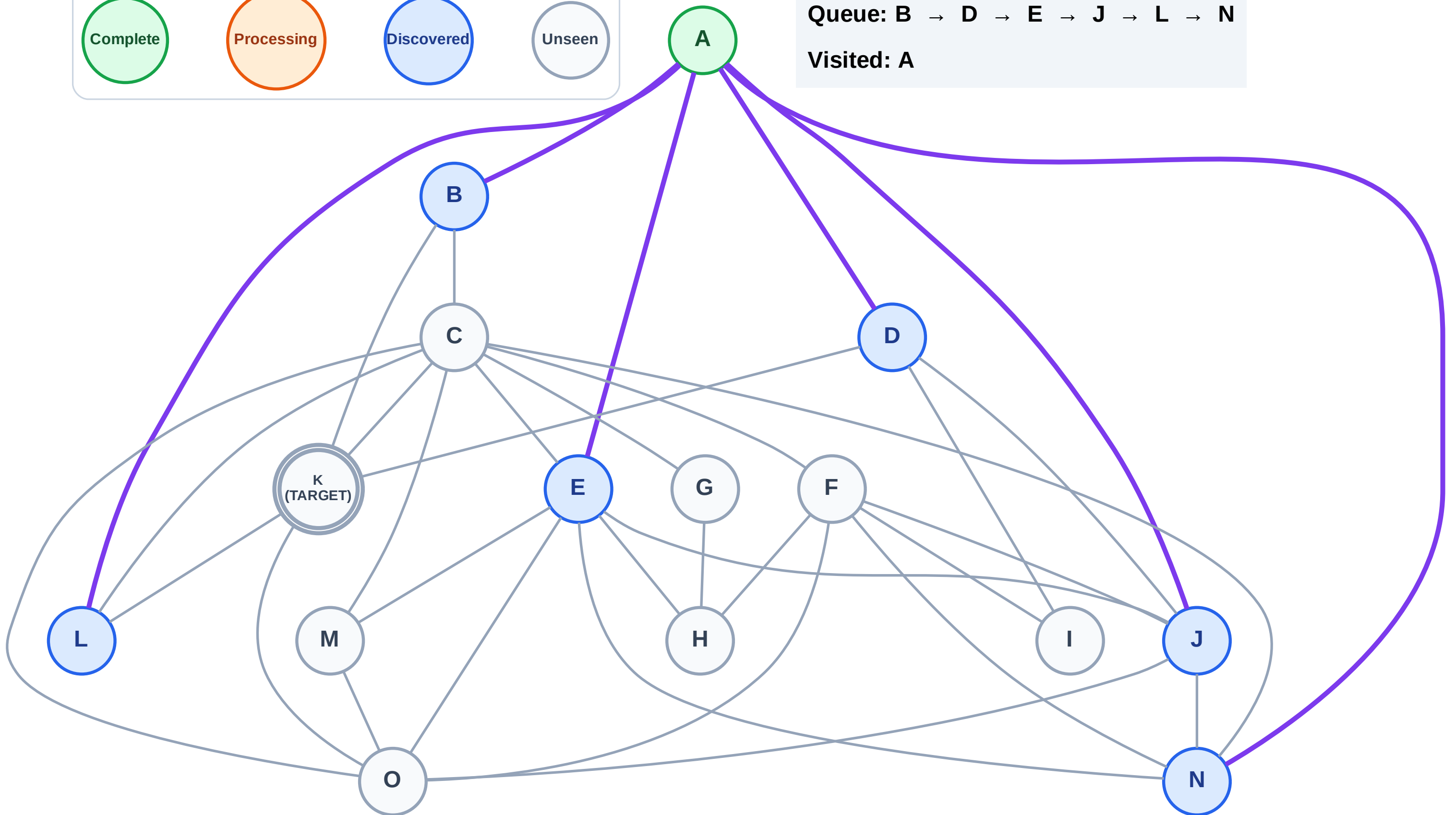
Step 3: Discover B, D, E, J, L, N from A

Legend



Queue: B → D → E → J → L → N

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

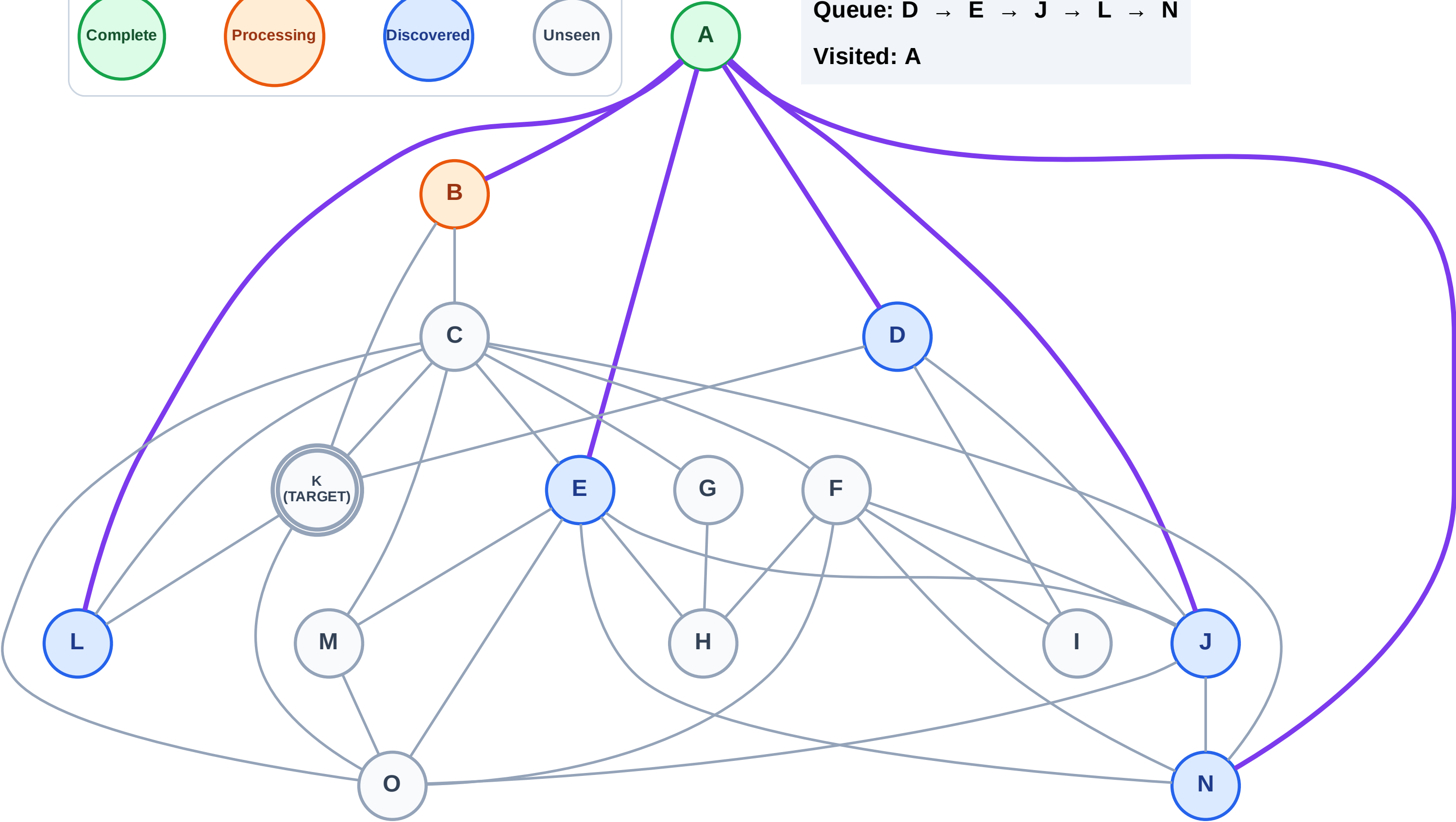
Processing

Discovered

Unseen

Queue: D → E → J → L → N

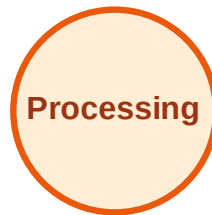
Visited: A



BFS Traversal

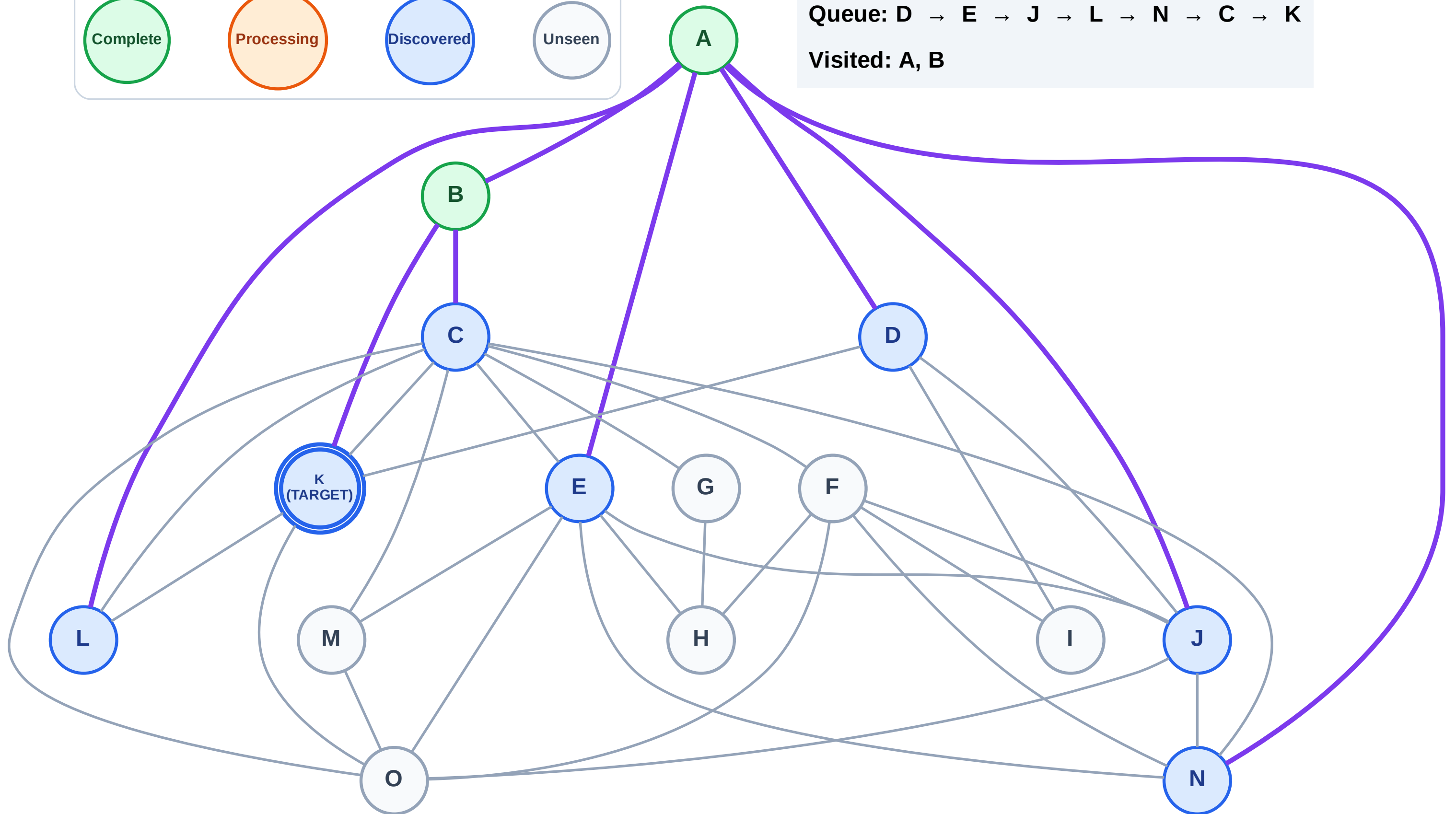
Step 5: Discover C, K from B

Legend



Queue: D → E → J → L → N → C → K

Visited: A, B



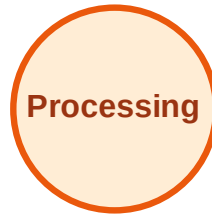
BFS Traversal

Step 6: Process node D

Legend



Complete



Processing



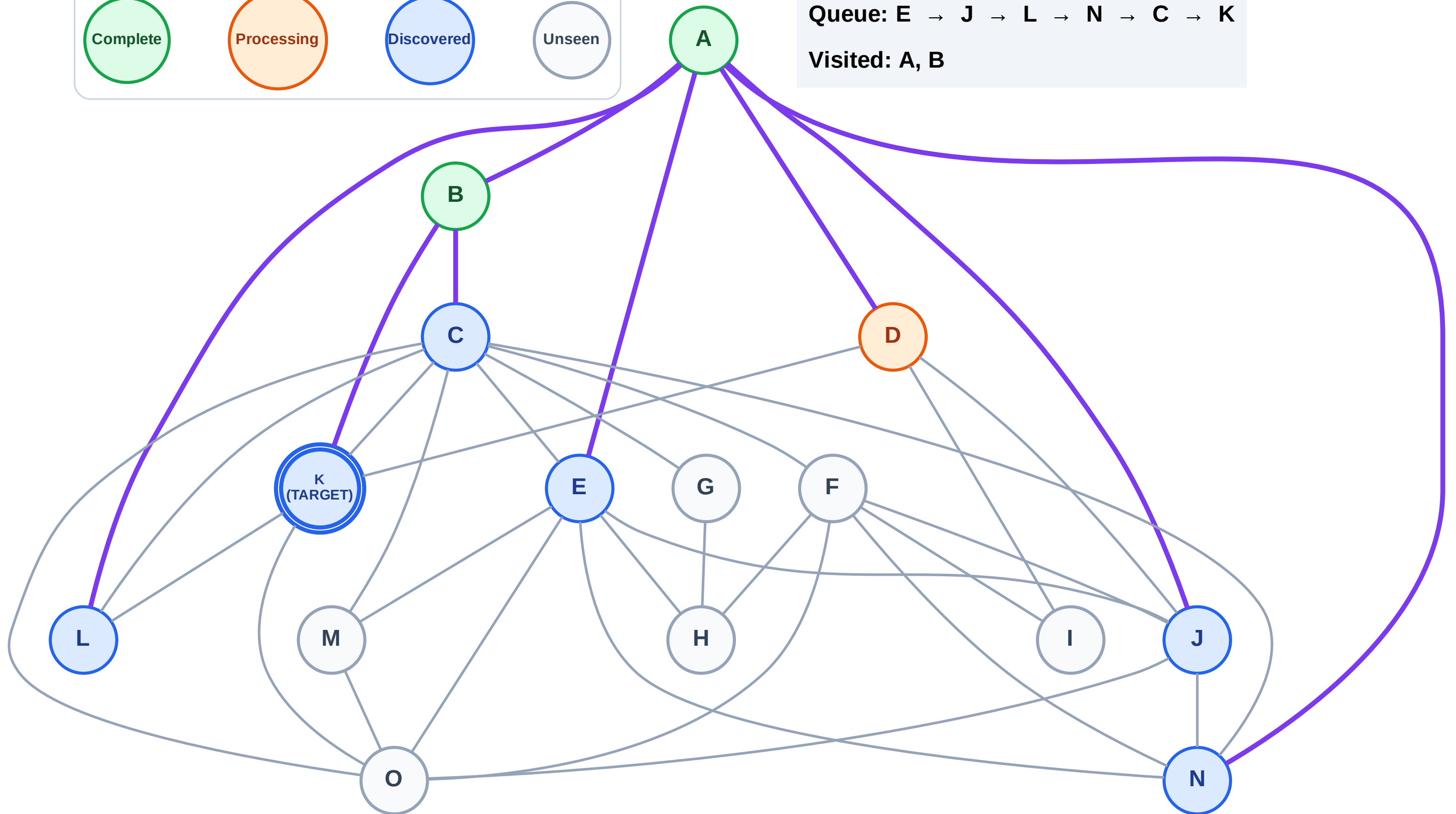
Discovered



Unseen

Queue: E → J → L → N → C → K

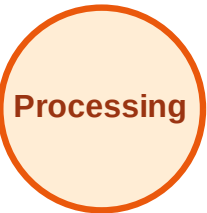
Visited: A, B



BFS Traversal

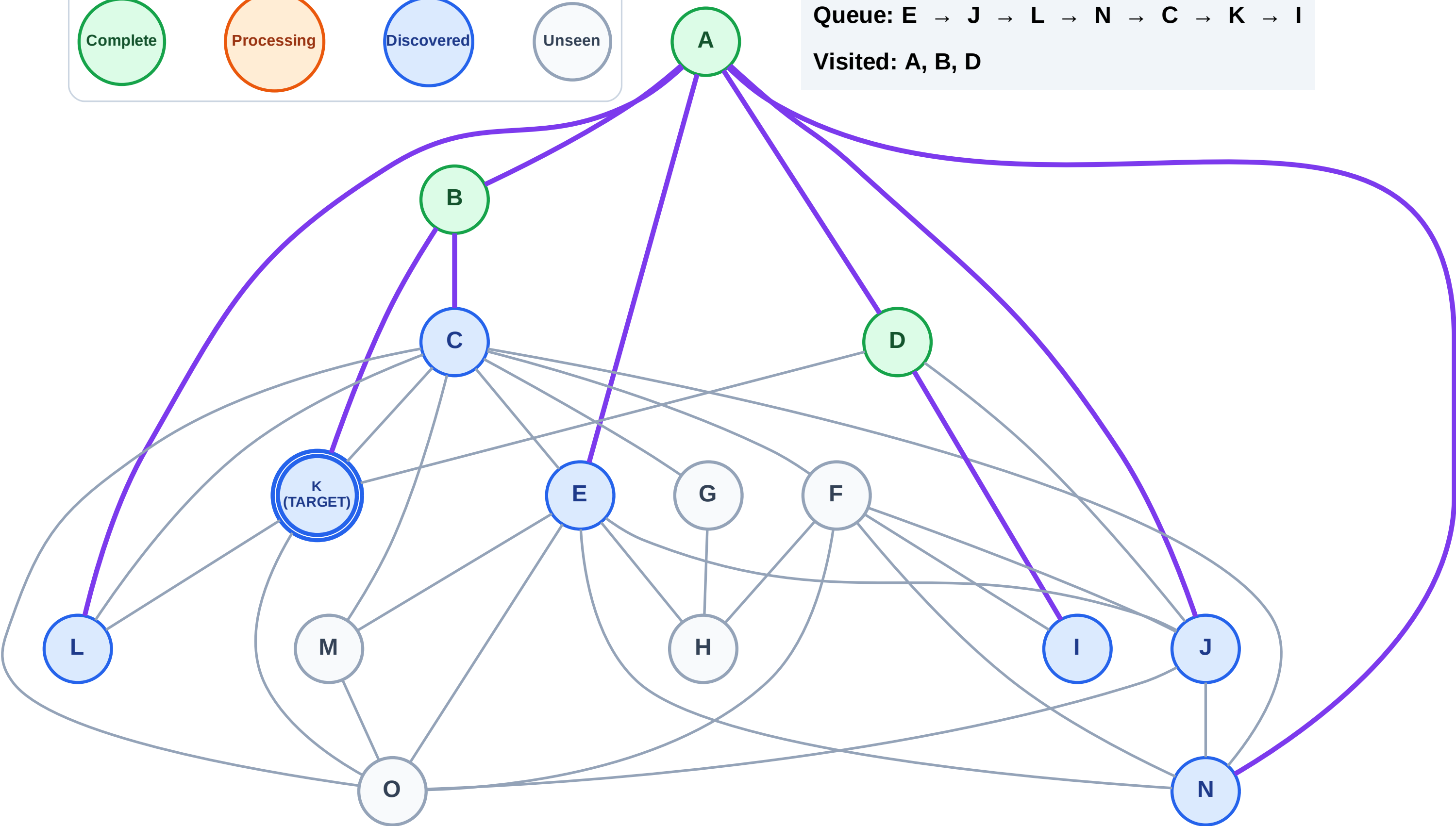
Step 7: Discover I from D

Legend



Queue: E → J → L → N → C → K → I

Visited: A, B, D



BFS Traversal

Step 8: Process node E

Legend

Complete

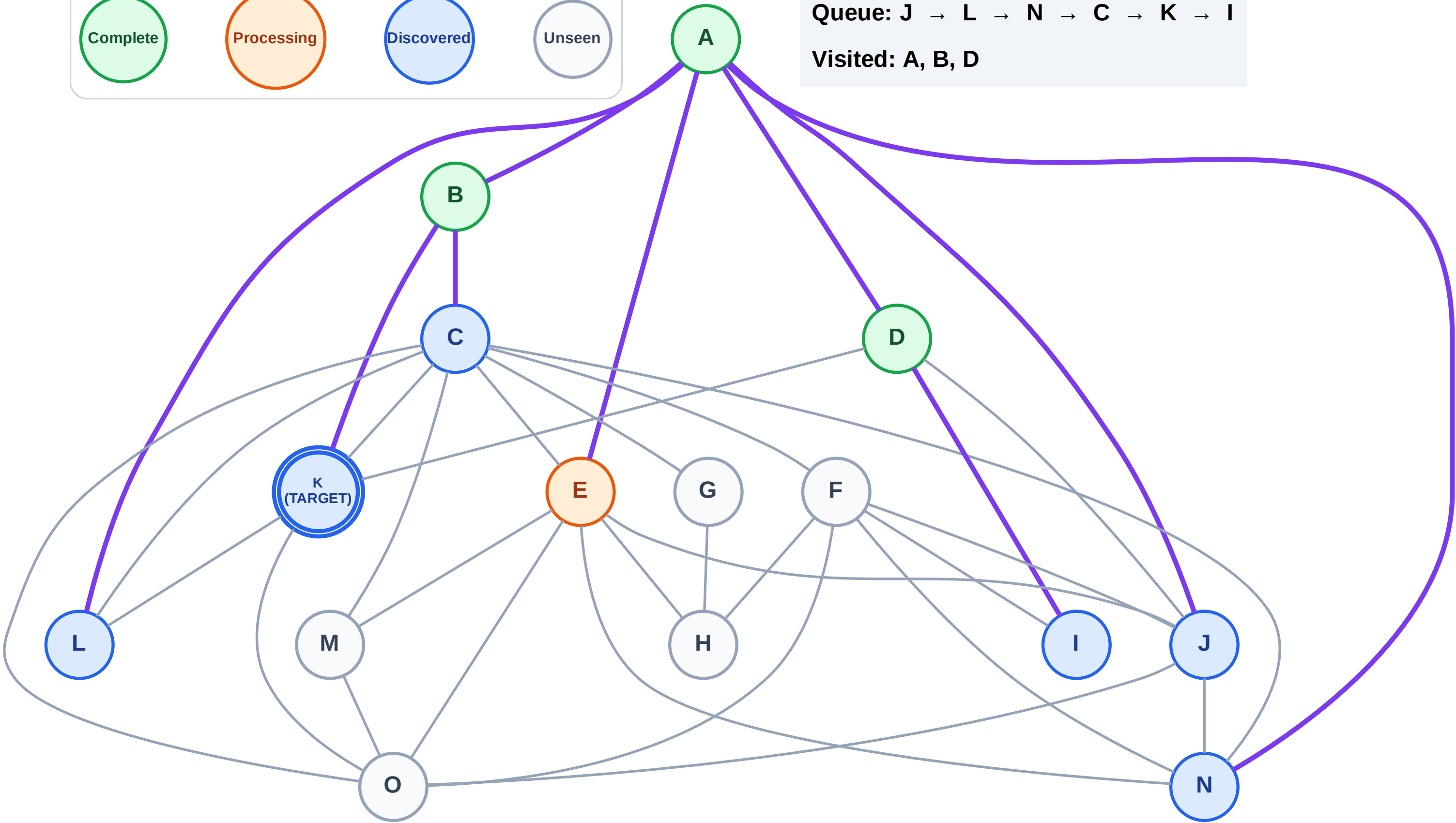
Processing

Discovered

Unseen

Queue: J → L → N → C → K → I

Visited: A, B, D



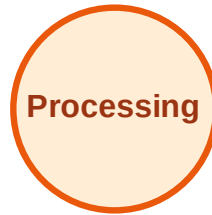
BFS Traversal

Step 9: Discover H, M, O from E

Legend



Complete



Processing



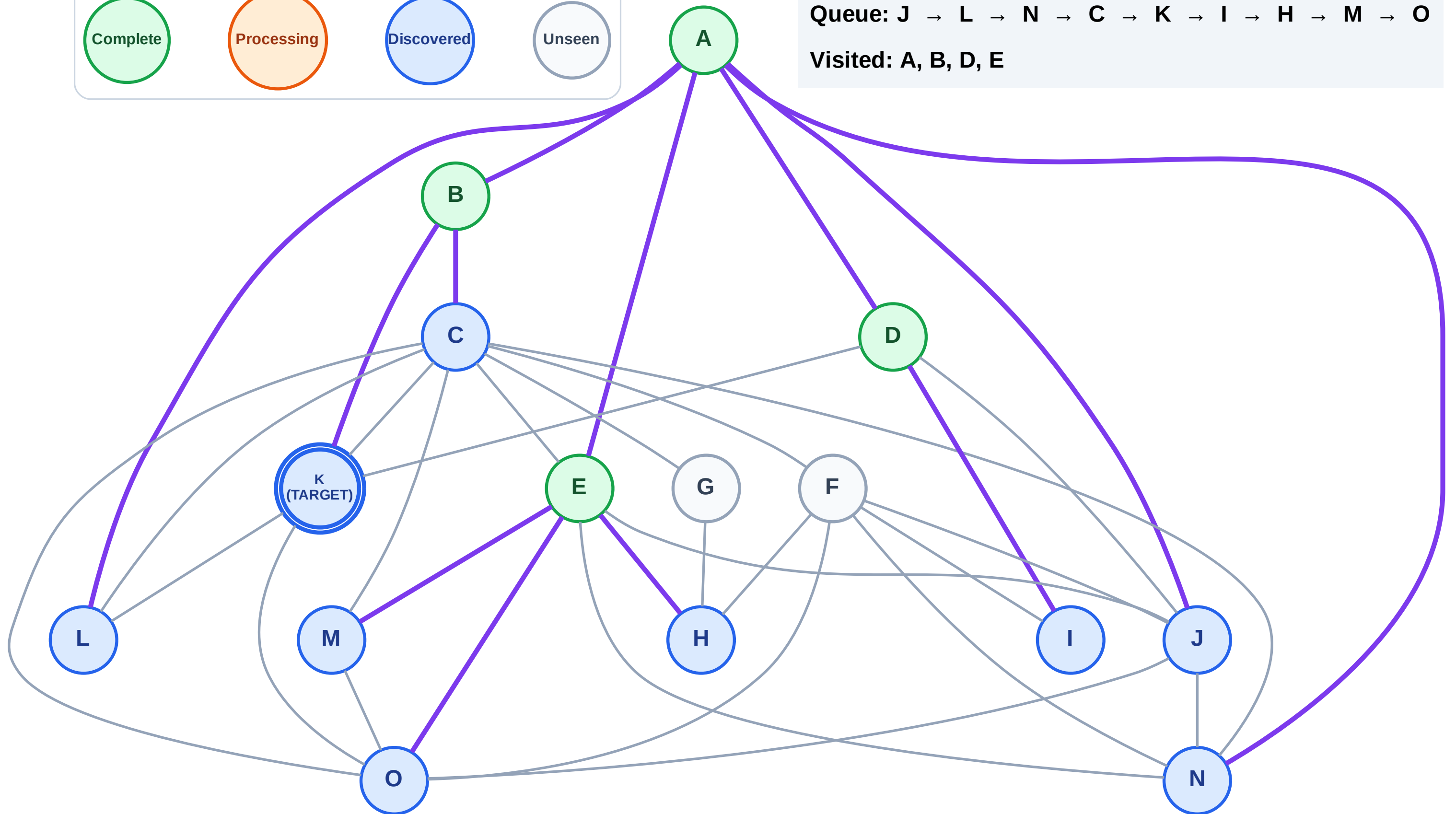
Discovered



Unseen

Queue: J → L → N → C → K → I → H → M → O

Visited: A, B, D, E



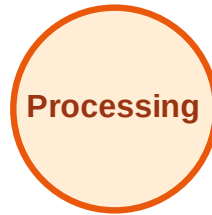
BFS Traversal

Step 10: Process node J

Legend



Complete



Processing



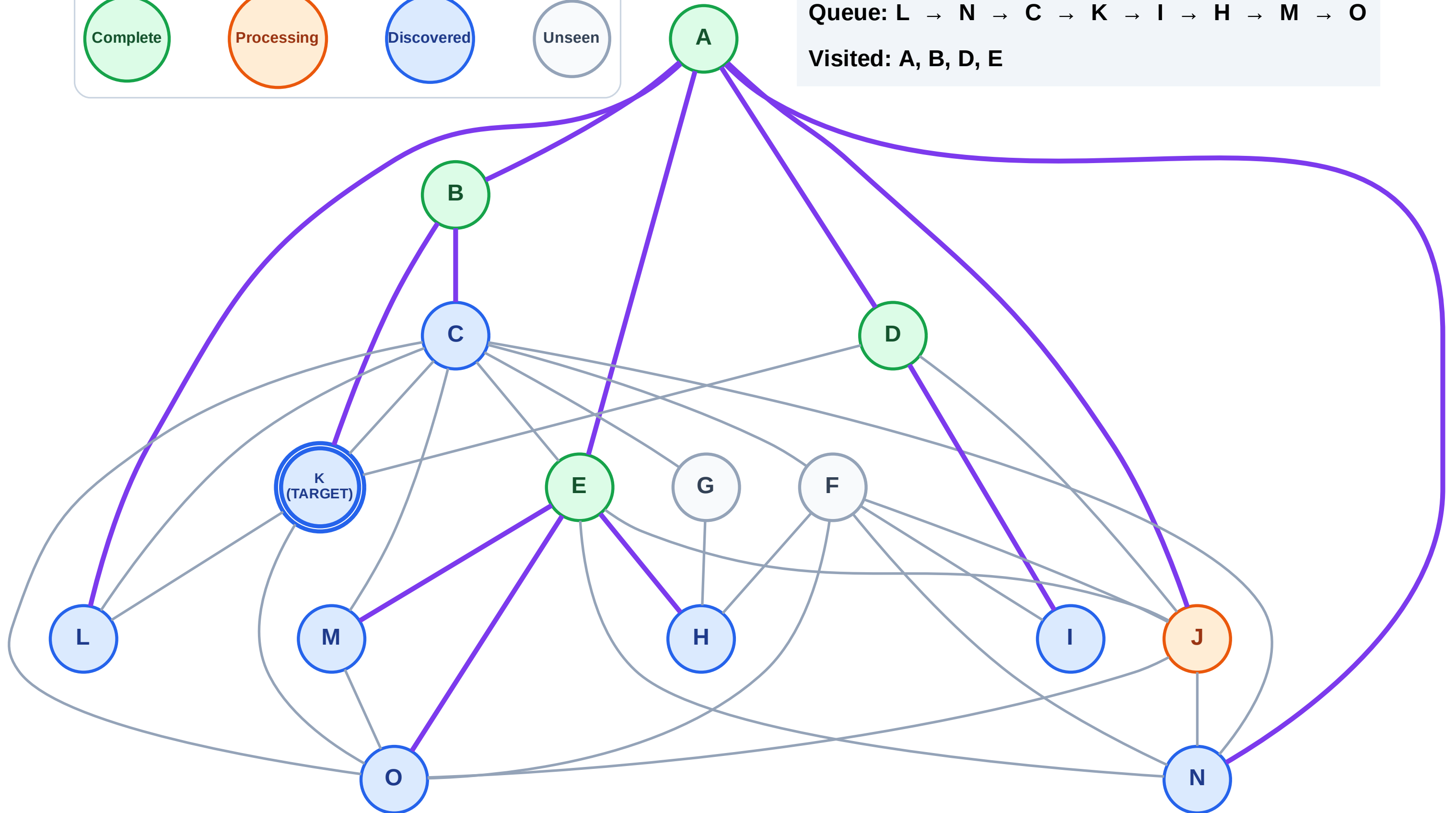
Discovered



Unseen

Queue: L → N → C → K → I → H → M → O

Visited: A, B, D, E



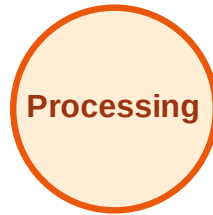
BFS Traversal

Step 11: Discover F from J

Legend



Complete



Processing



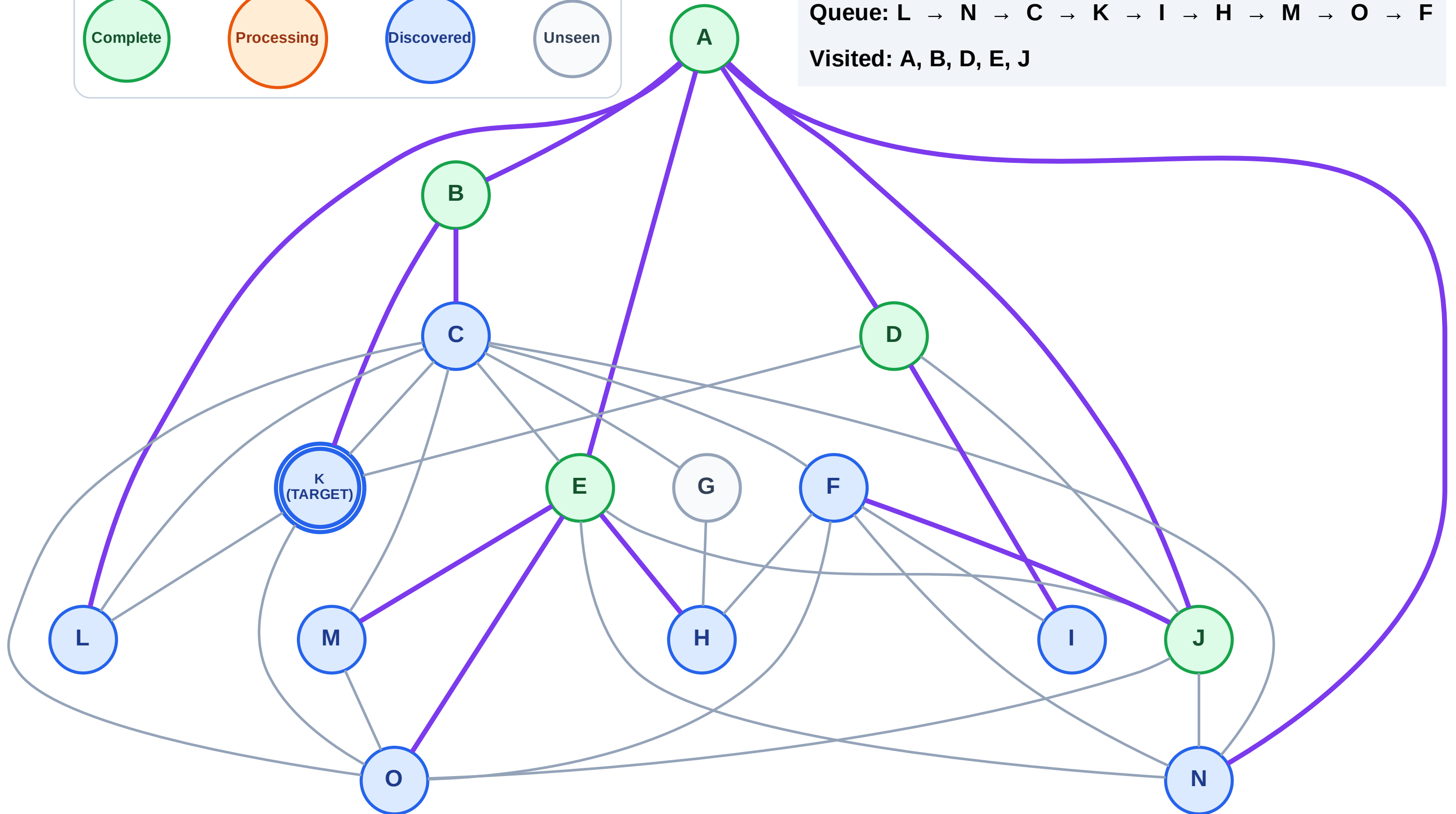
Discovered



Unseen

Queue: L → N → C → K → I → H → M → O → F

Visited: A, B, D, E, J



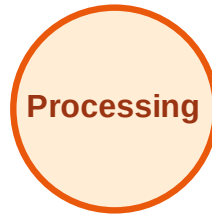
BFS Traversal

Step 12: Process node L

Legend



Complete



Processing



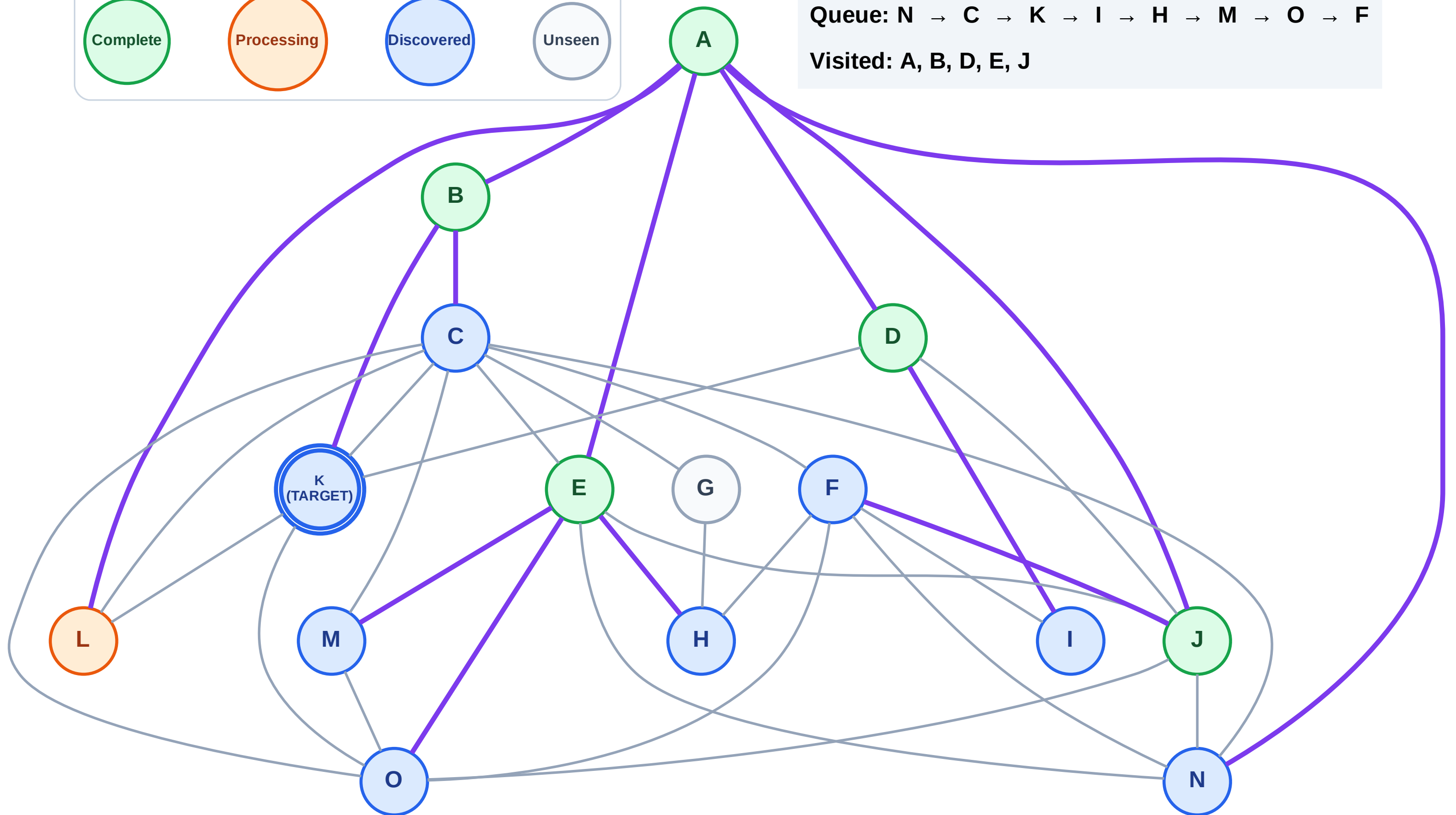
Discovered



Unseen

Queue: N → C → K → I → H → M → O → F

Visited: A, B, D, E, J



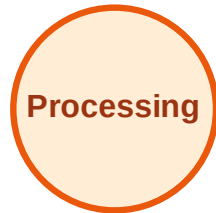
BFS Traversal

Step 13: No new neighbors from L

Legend



Complete



Processing



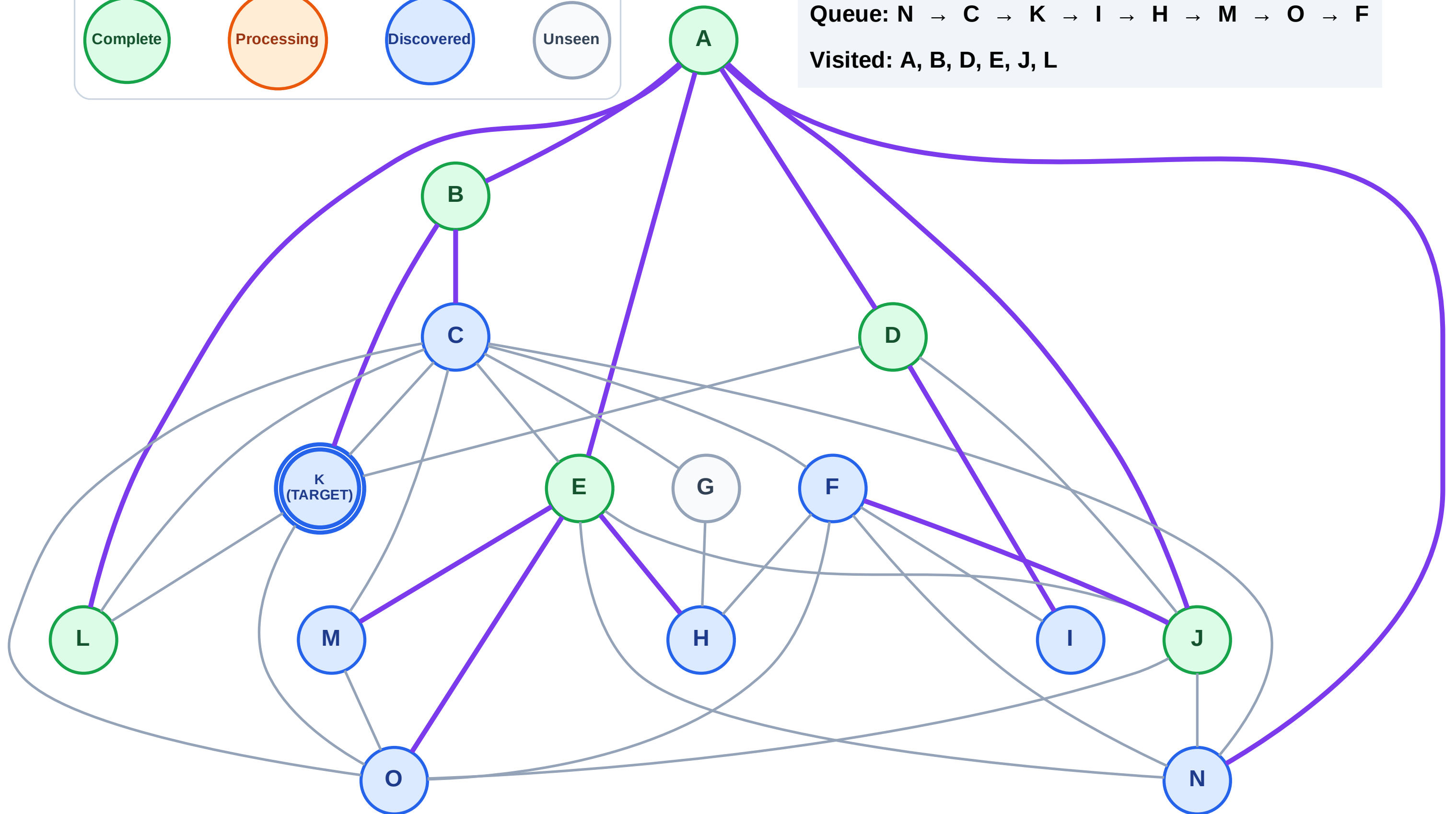
Discovered



Unseen

Queue: N → C → K → I → H → M → O → F

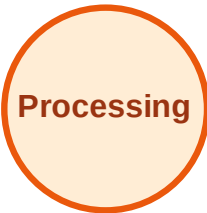
Visited: A, B, D, E, J, L



BFS Traversal

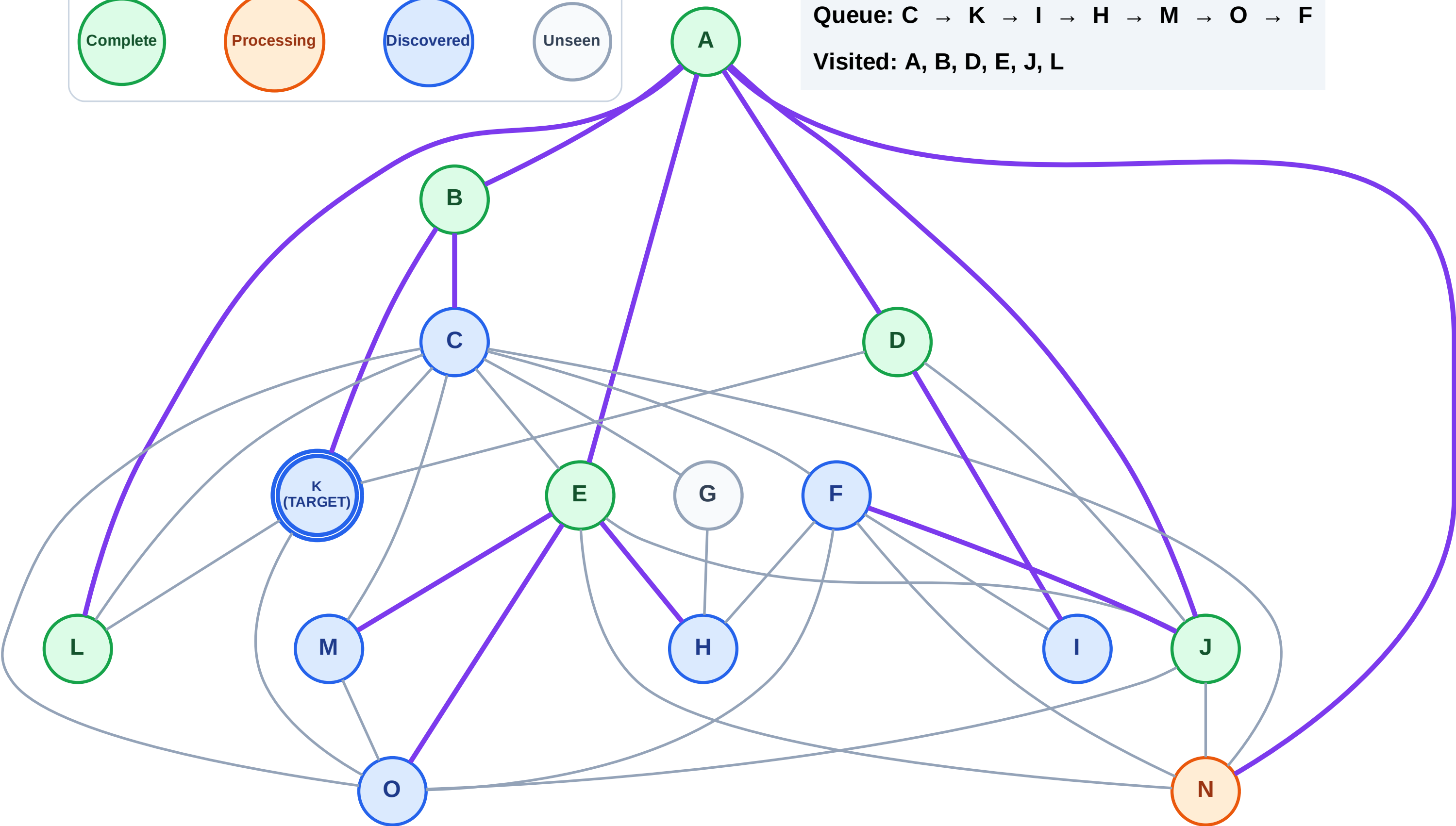
Step 14: Process node N

Legend



Queue: C → K → I → H → M → O → F

Visited: A, B, D, E, J, L



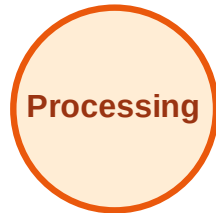
BFS Traversal

Step 15: No new neighbors from N

Legend



Complete



Processing



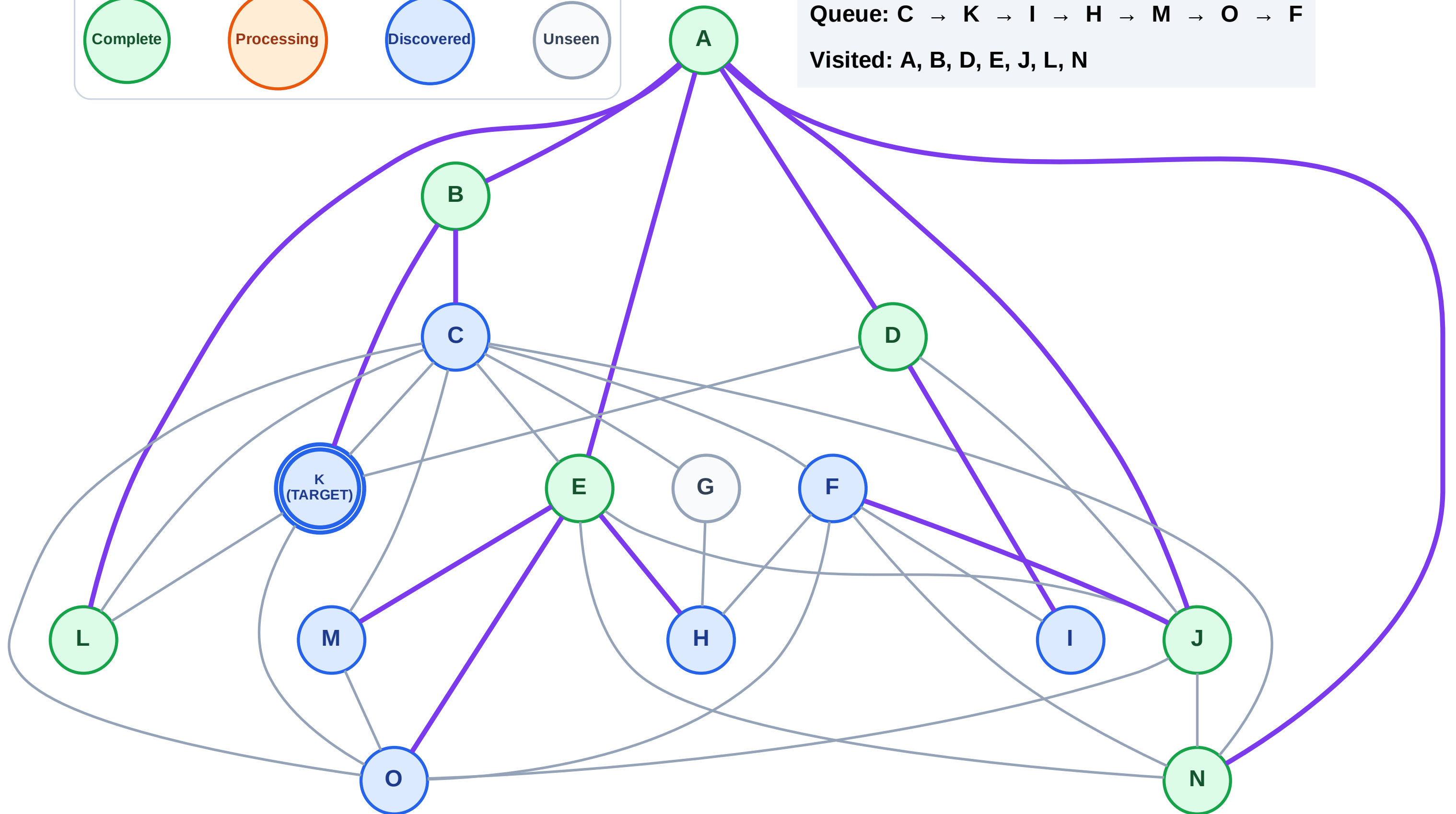
Discovered



Unseen

Queue: C → K → I → H → M → O → F

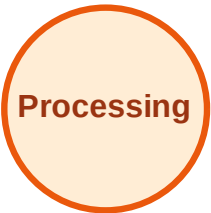
Visited: A, B, D, E, J, L, N



BFS Traversal

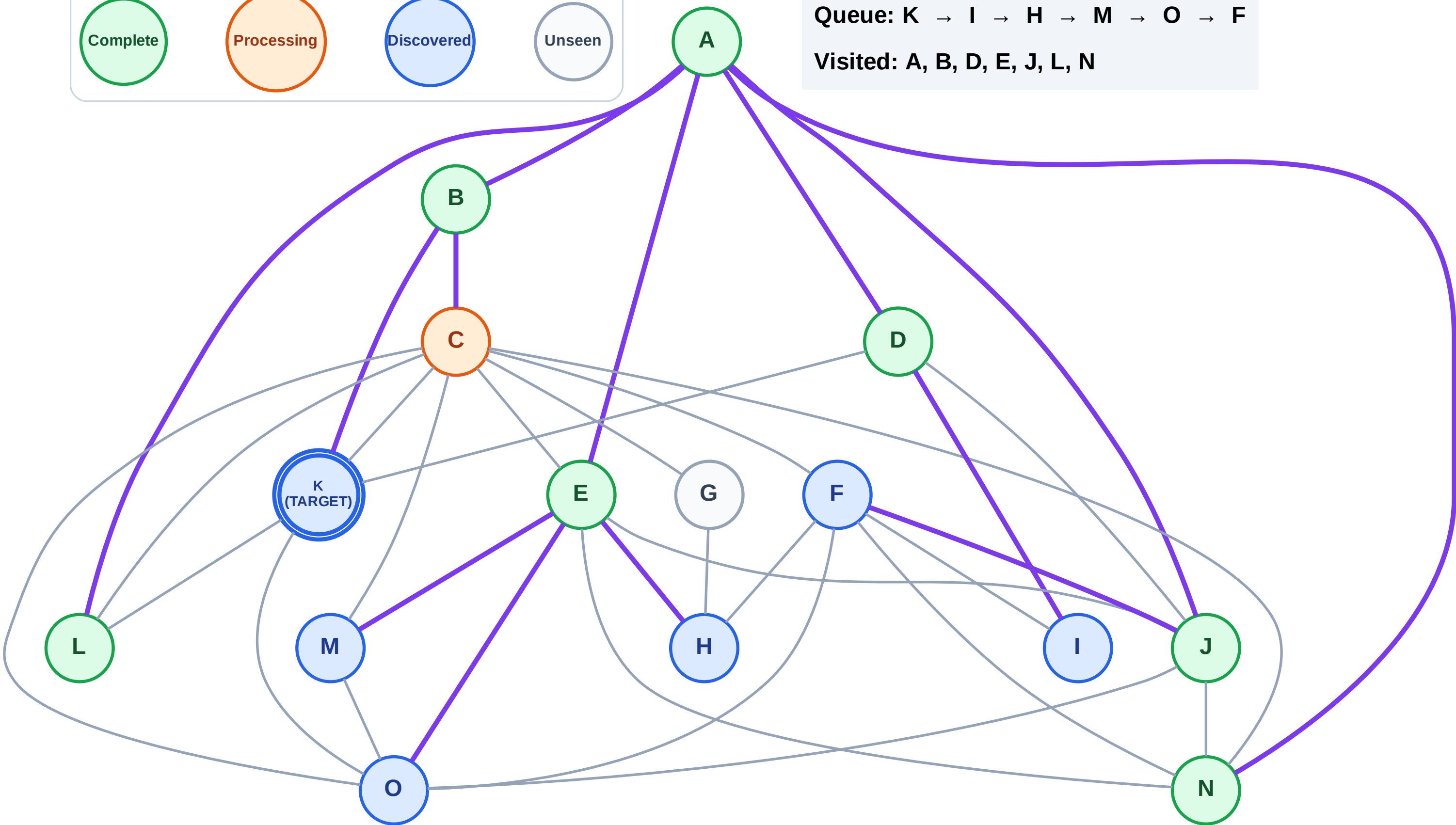
Step 16: Process node C

Legend



Queue: K → I → H → M → O → F

Visited: A, B, D, E, J, L, N



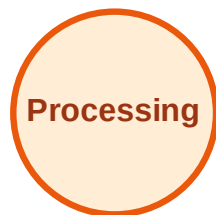
BFS Traversal

Step 17: Discover G from C

Legend



Complete



Processing



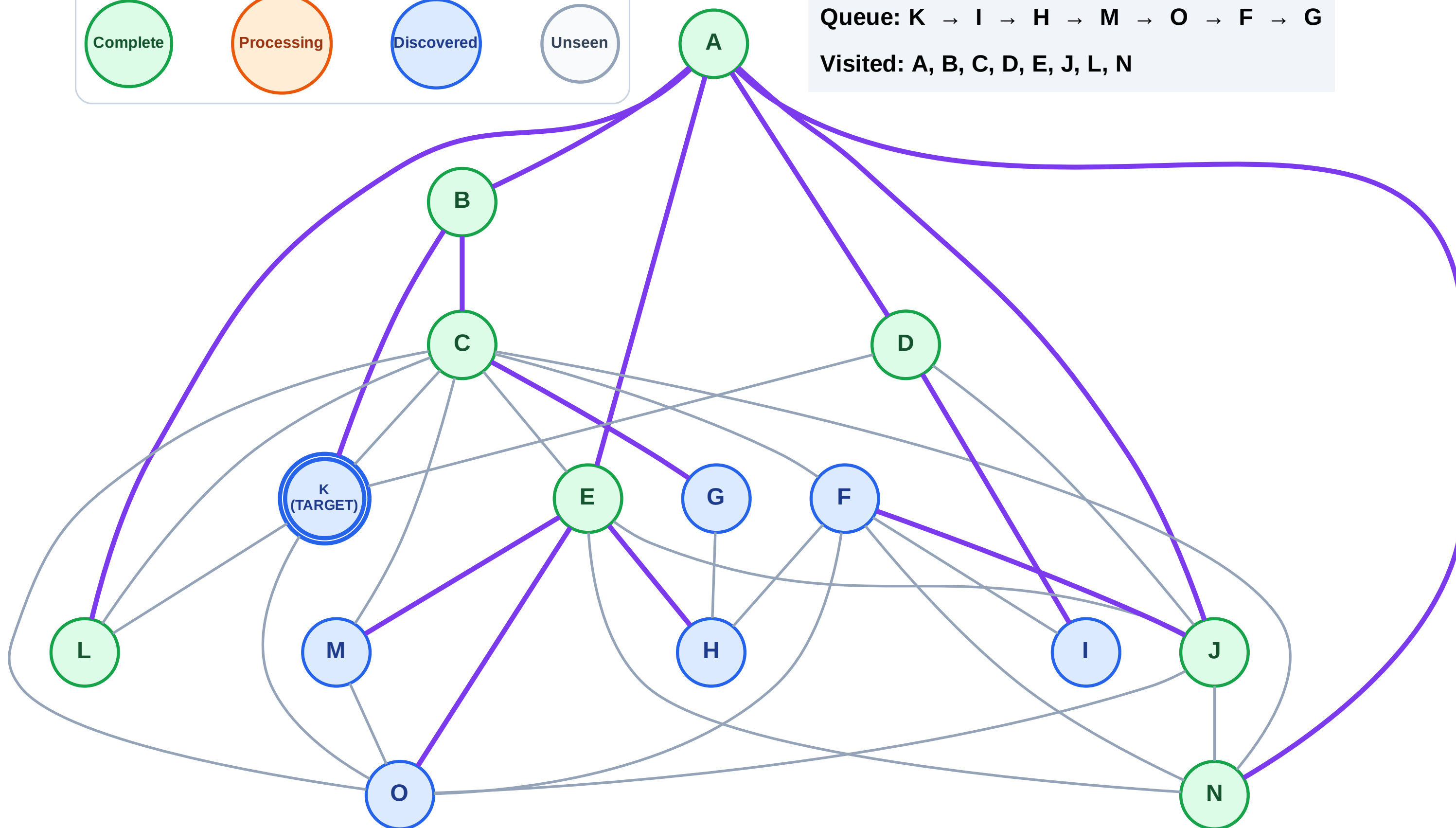
Discovered



Unseen

Queue: K → I → H → M → O → F → G

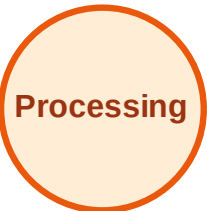
Visited: A, B, C, D, E, J, L, N



BFS Traversal

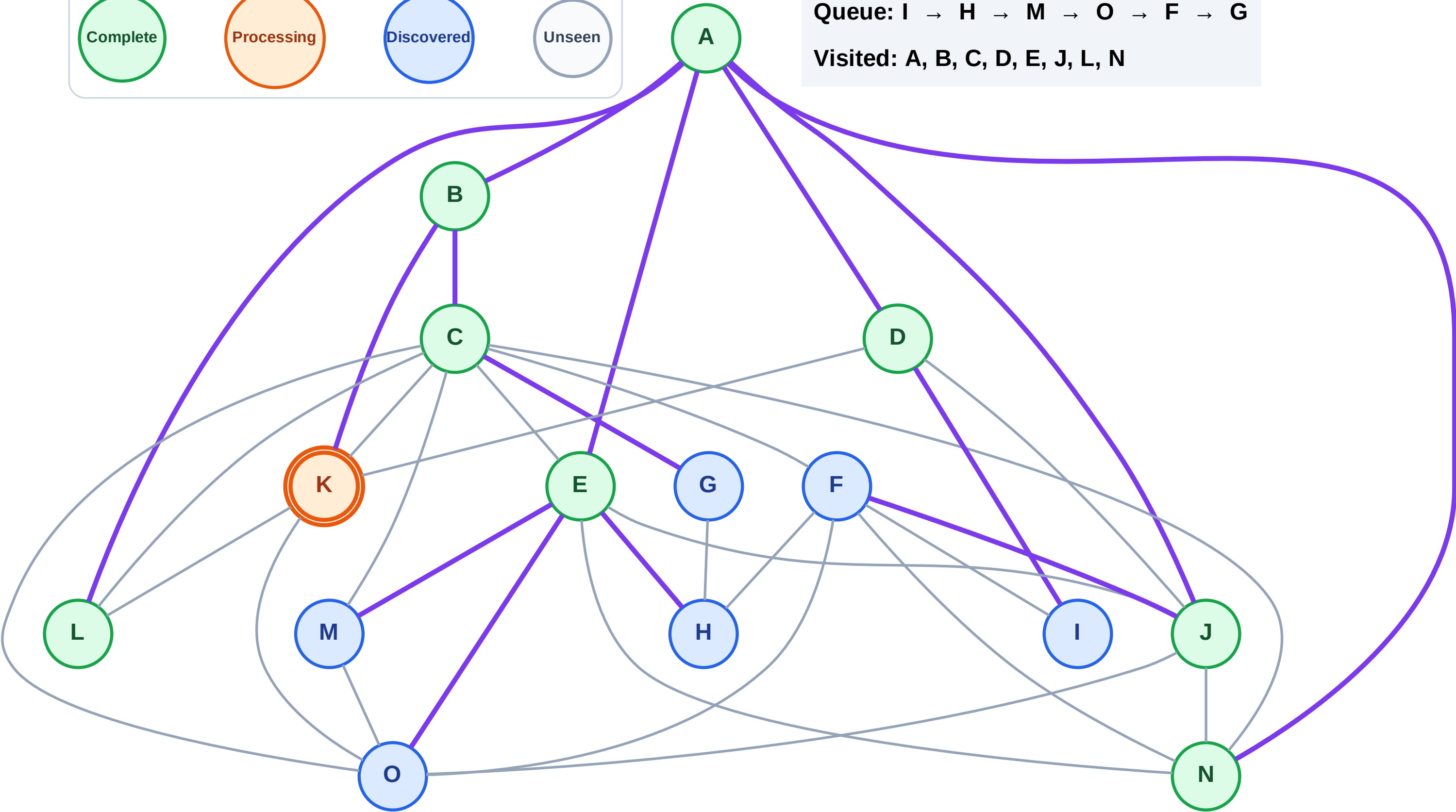
Step 18: Process node K

Legend



Queue: I → H → M → O → F → G

Visited: A, B, C, D, E, J, L, N



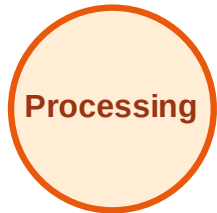
BFS Traversal

Step 19: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Complete



Processing



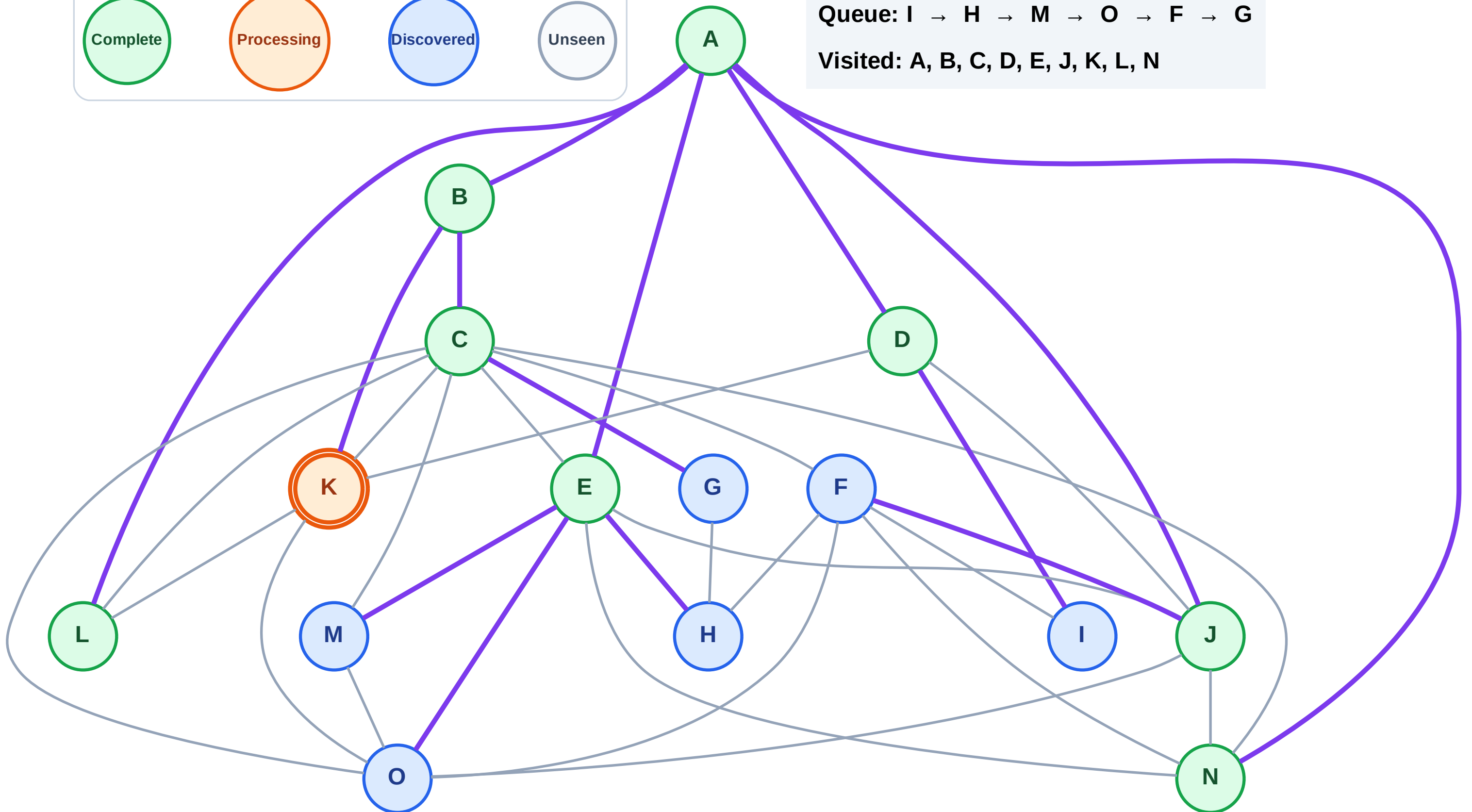
Discovered



Unseen

Queue: I → H → M → O → F → G

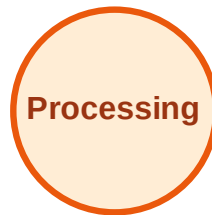
Visited: A, B, C, D, E, J, K, L, N



DFS Traversal

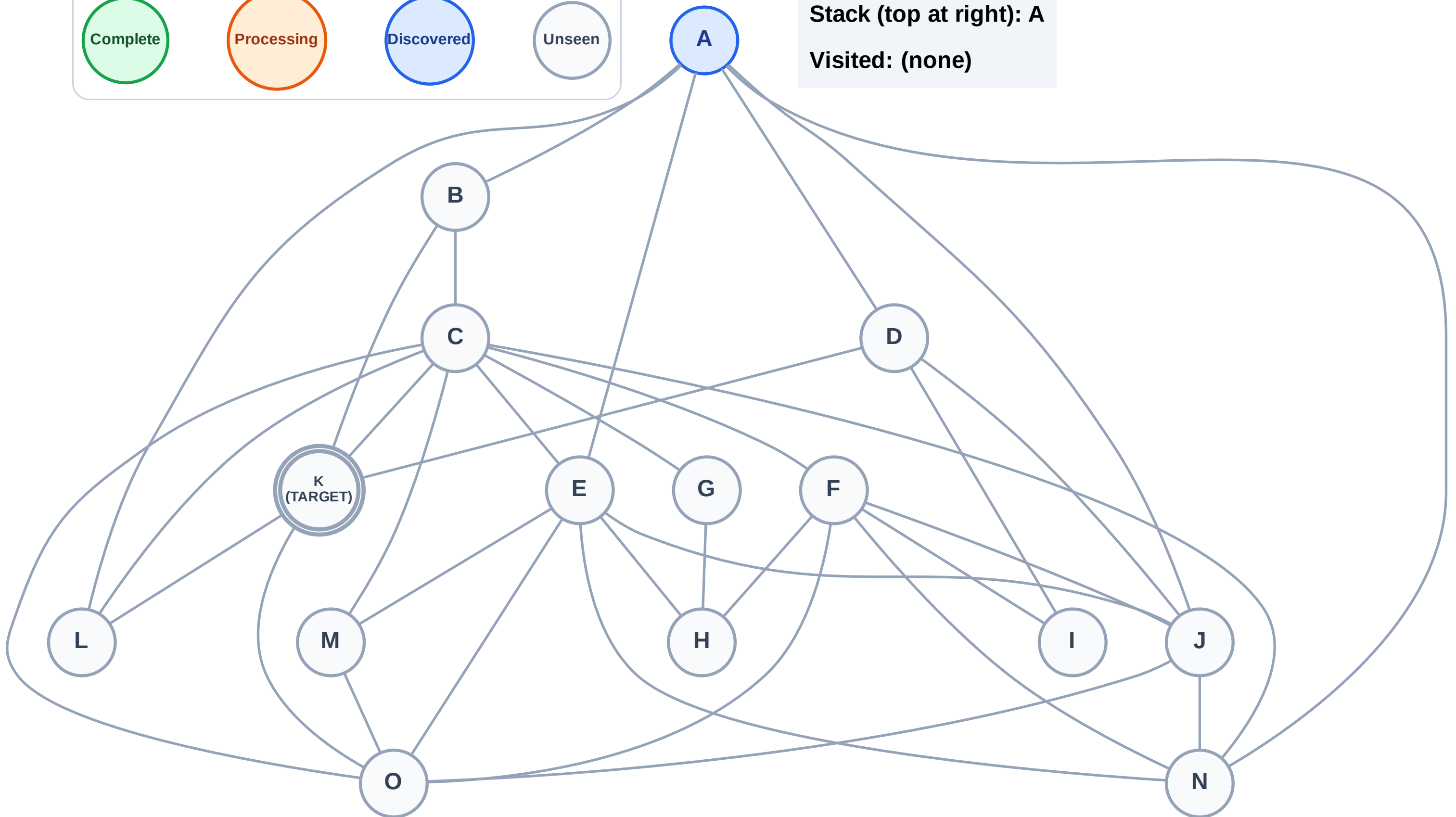
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

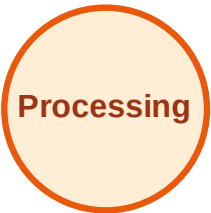
Visited: (none)



DFS Traversal

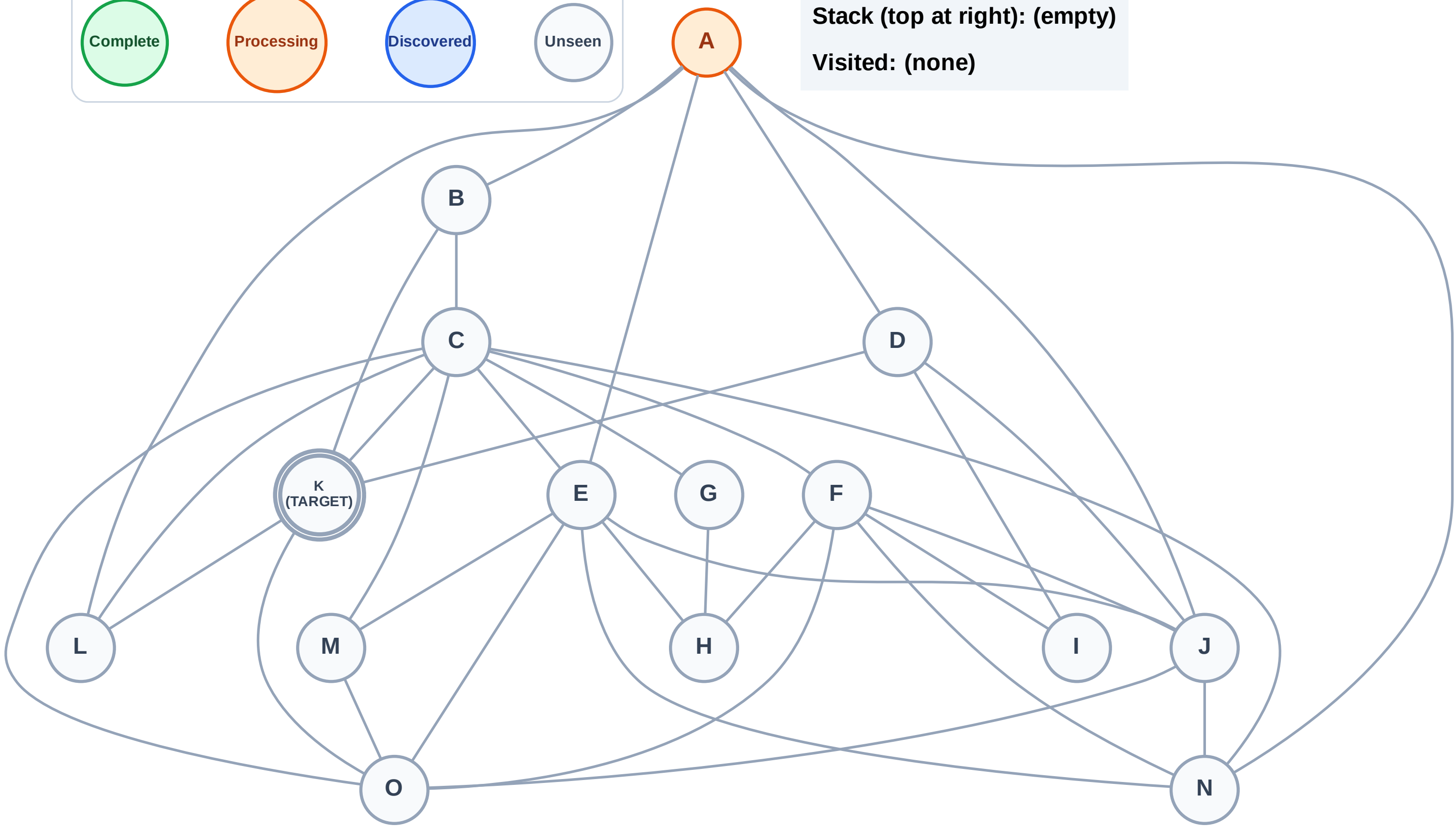
Step 2: Process node A

Legend



Stack (top at right): (empty)

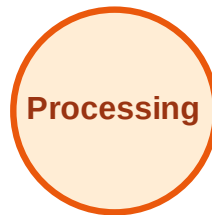
Visited: (none)



DFS Traversal

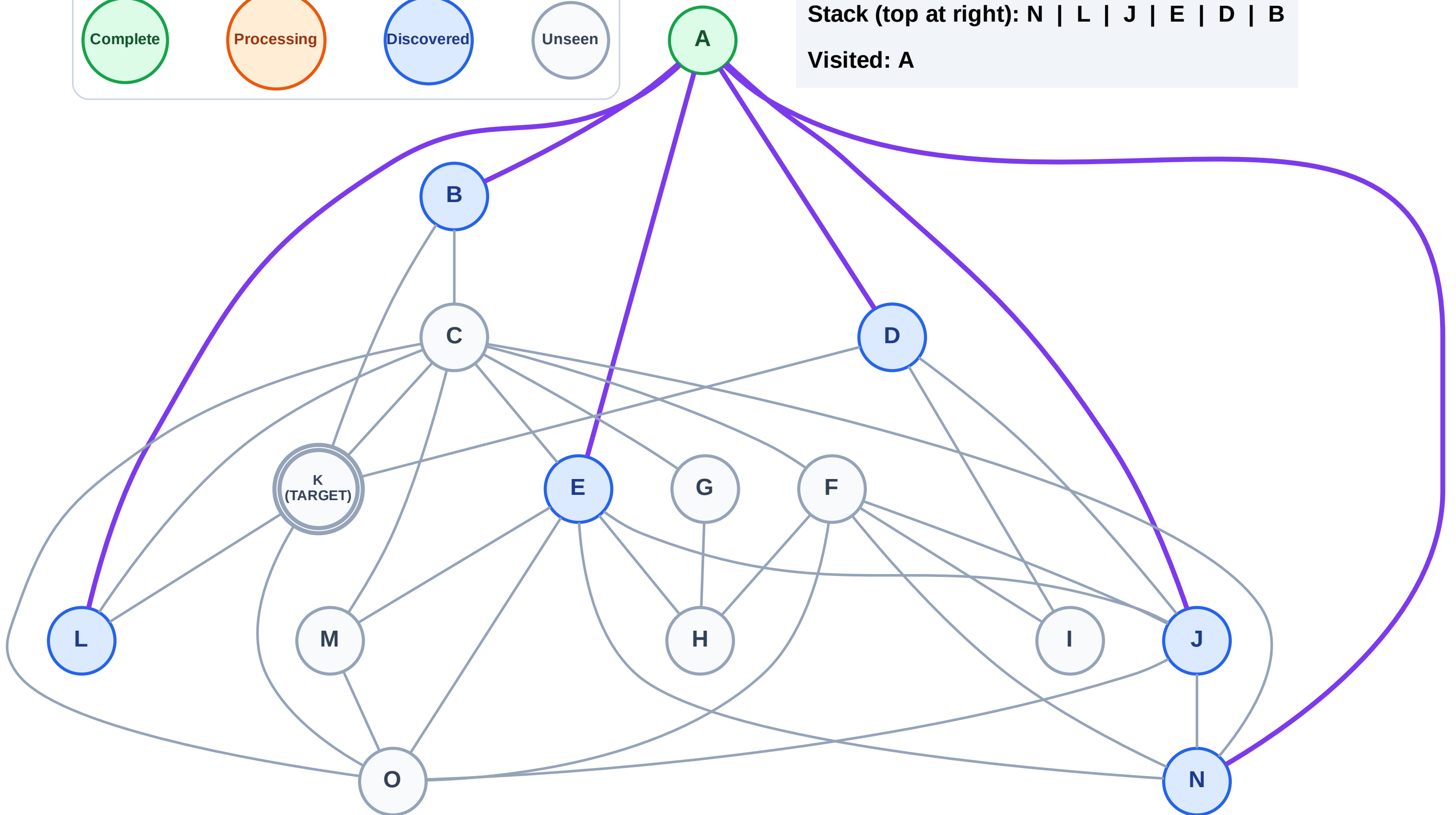
Step 3: Discover N, L, J, E, D, B from A

Legend



Stack (top at right): N | L | J | E | D | B

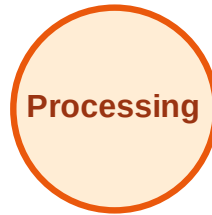
Visited: A



DFS Traversal

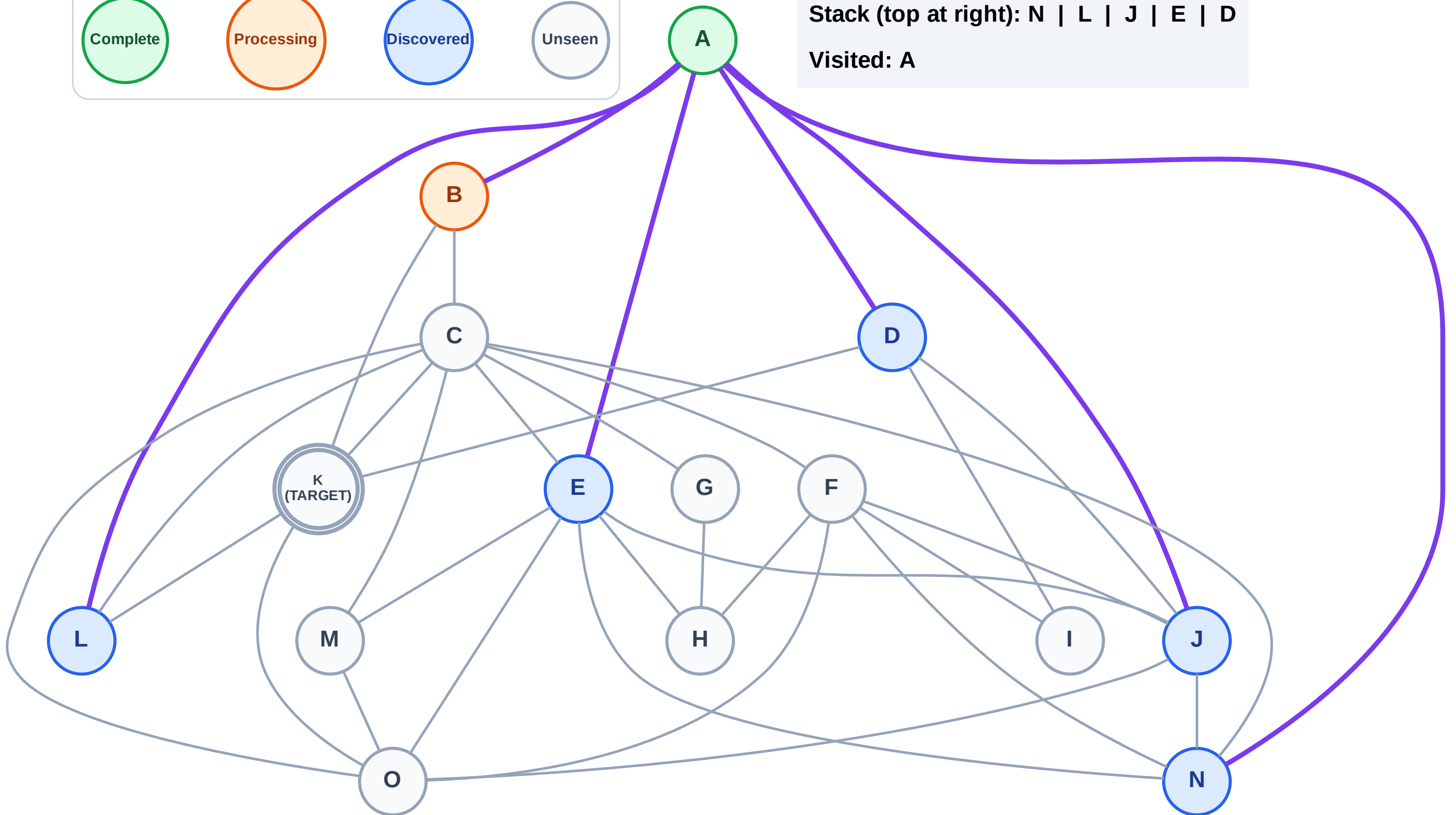
Step 4: Process node B

Legend



Stack (top at right): N | L | J | E | D

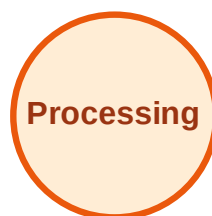
Visited: A



DFS Traversal

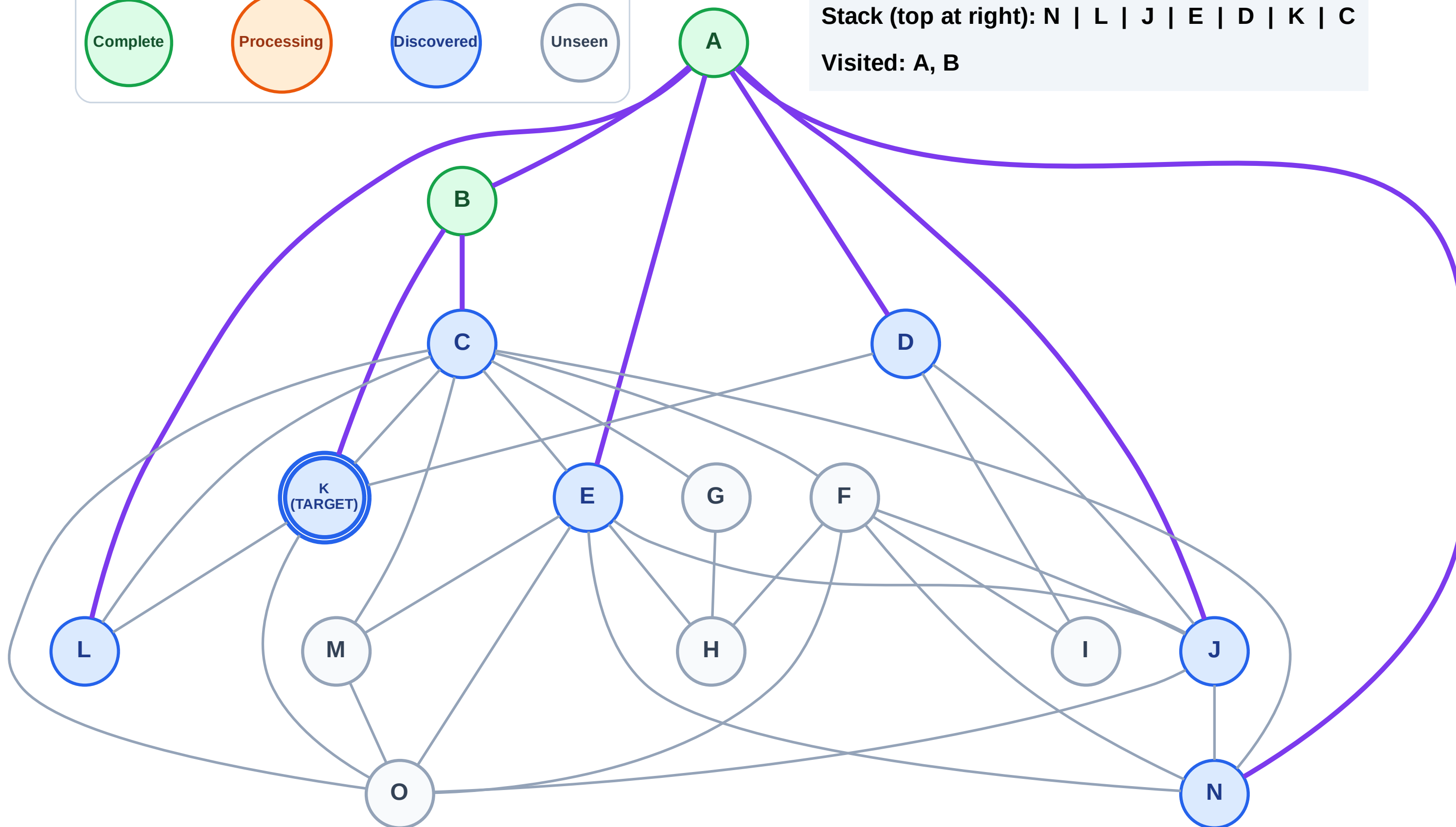
Step 5: Discover K, C from B

Legend



Stack (top at right): N | L | J | E | D | K | C

Visited: A, B



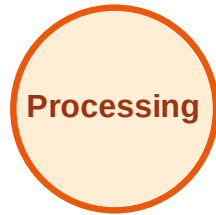
DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



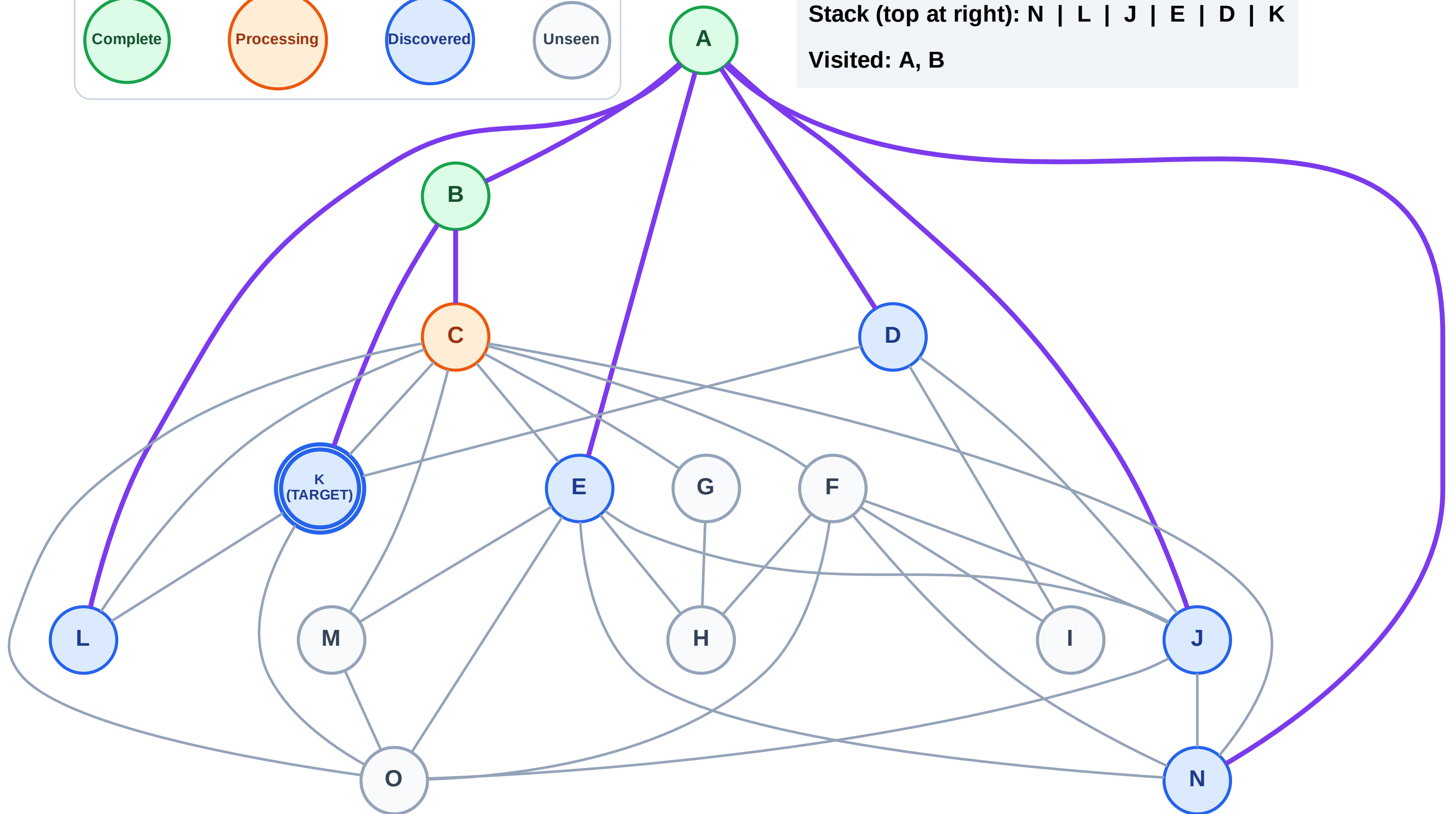
Discovered



Unseen

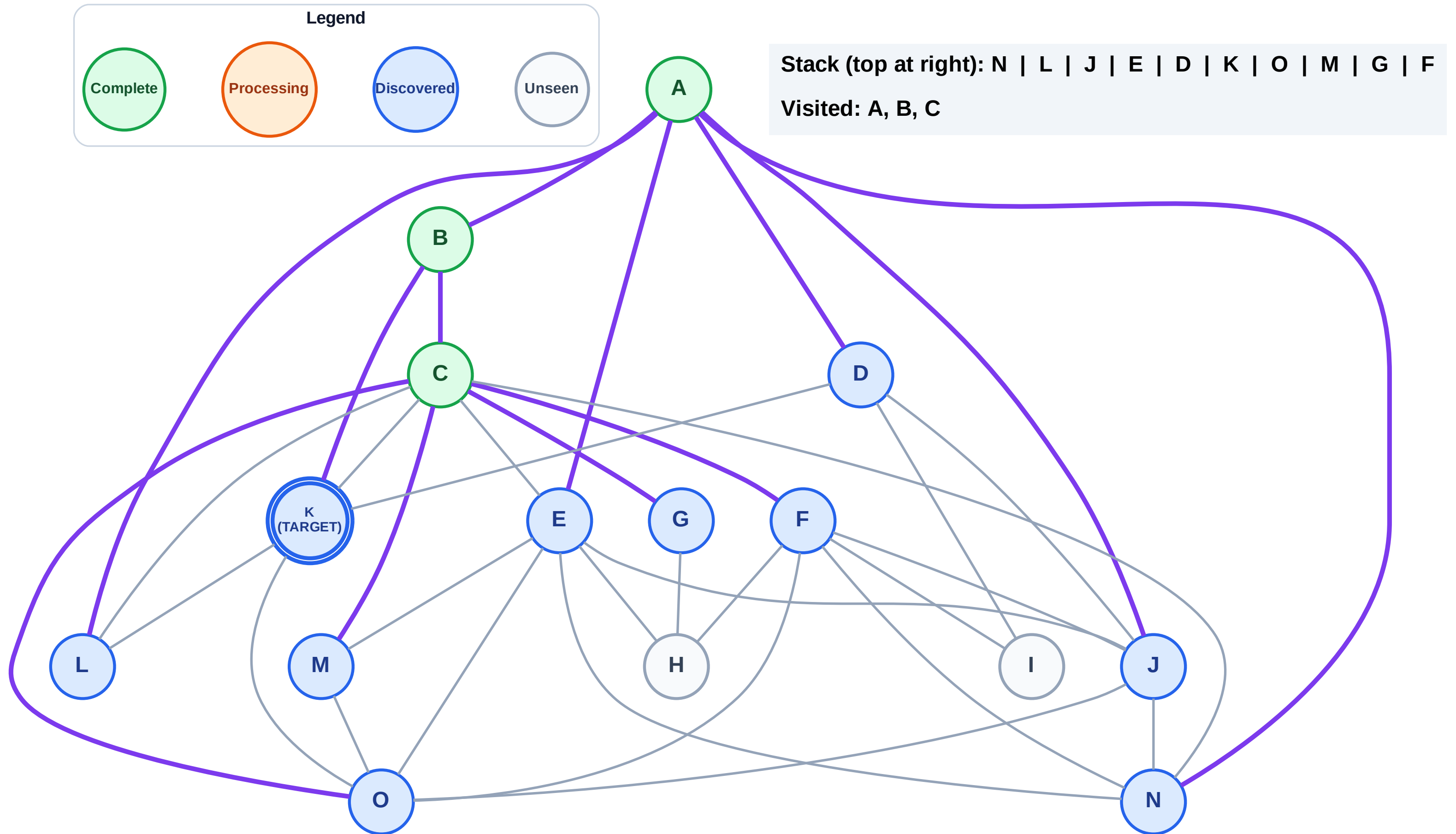
Stack (top at right): N | L | J | E | D | K

Visited: A, B



Step 7: Discover O, M, G, F from C

Step 7: Discover O, M, G, F from C



DFS Traversal

Step 8: Process node F

Legend

Complete

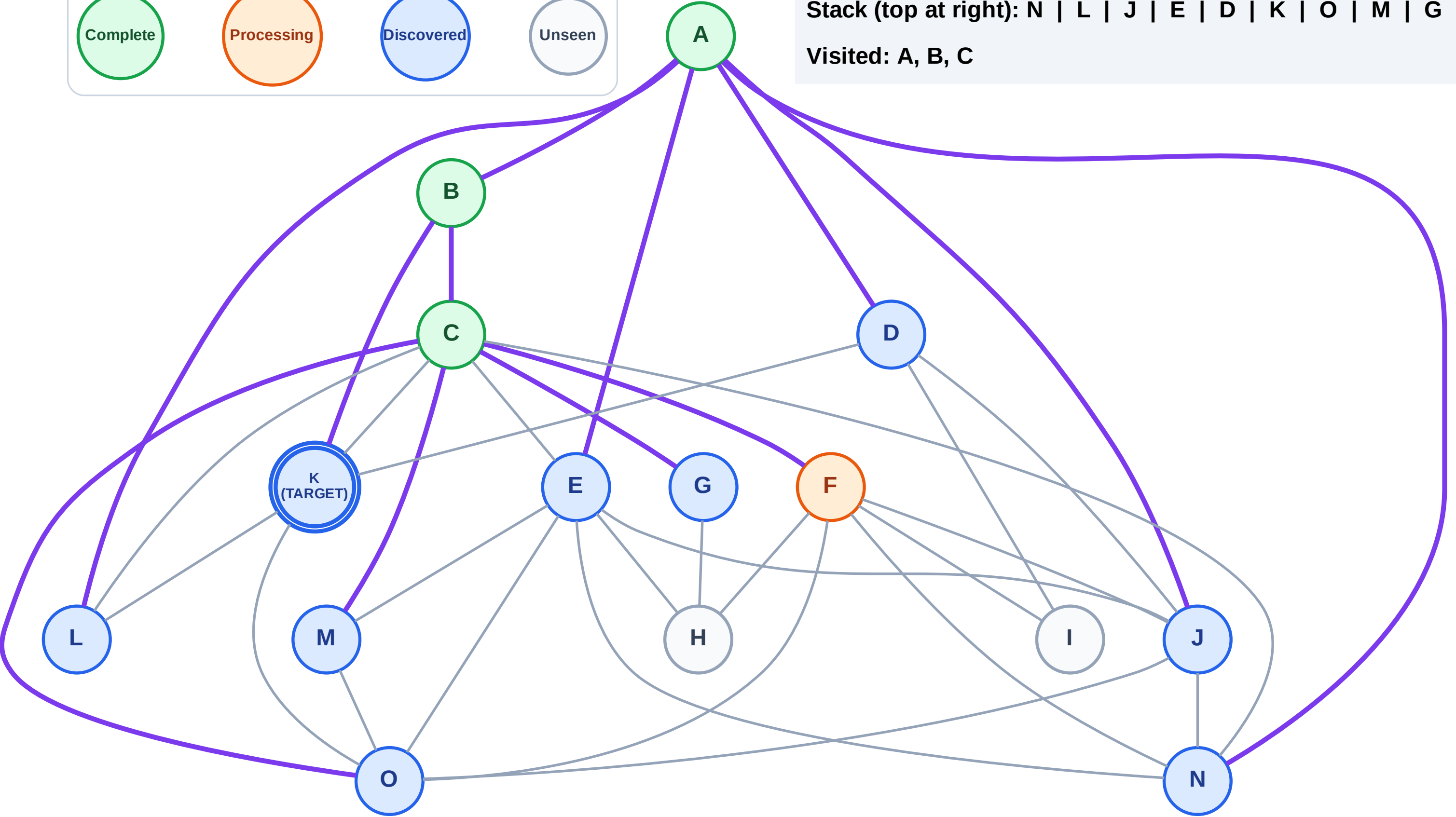
Processing

Discovered

Unseen

Stack (top at right): N | L | J | E | D | K | O | M | G

Visited: A, B, C



DFS Traversal

Step 9: Discover I, H from F

Legend

Complete

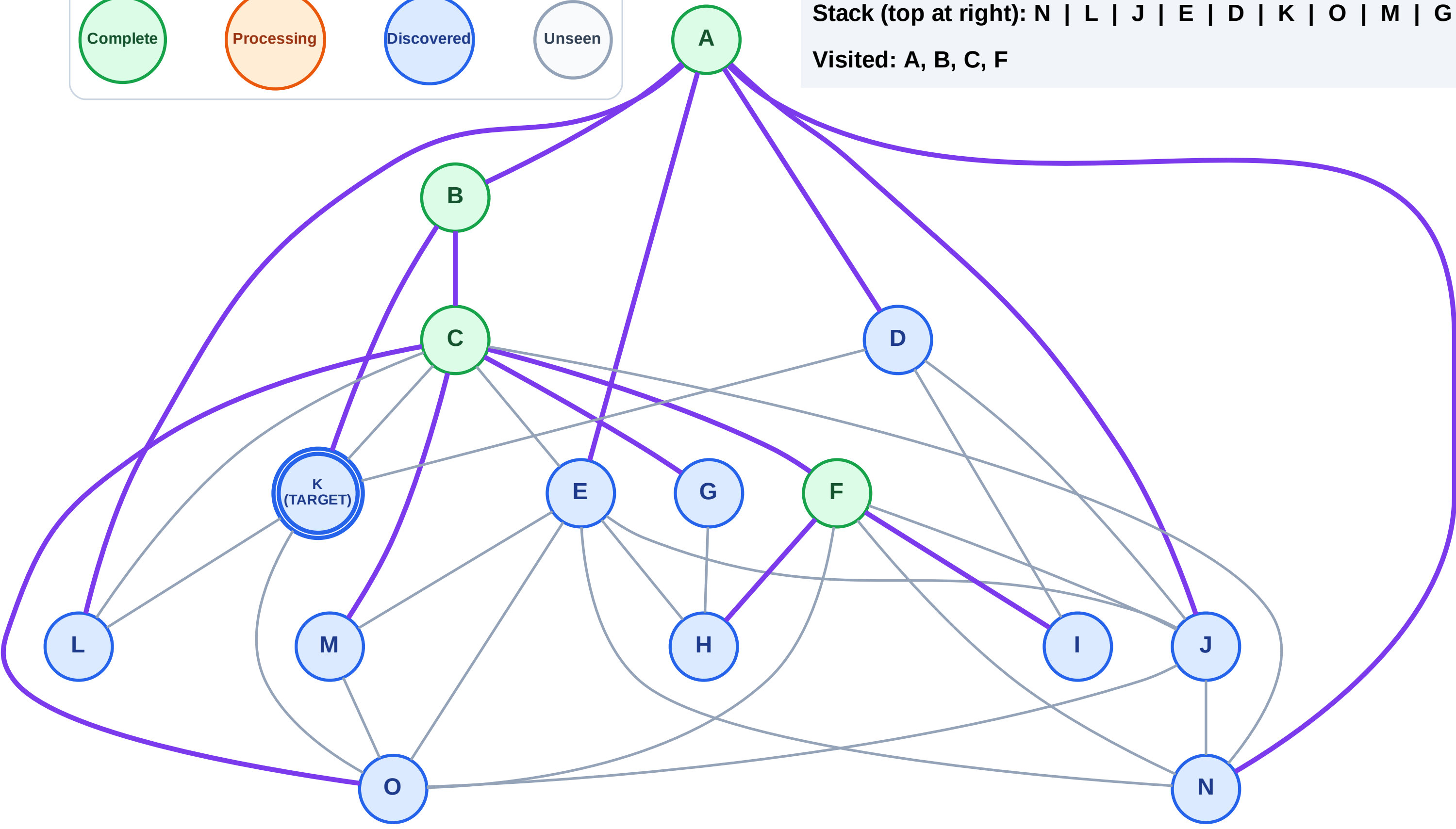
Processing

Discovered

Unseen

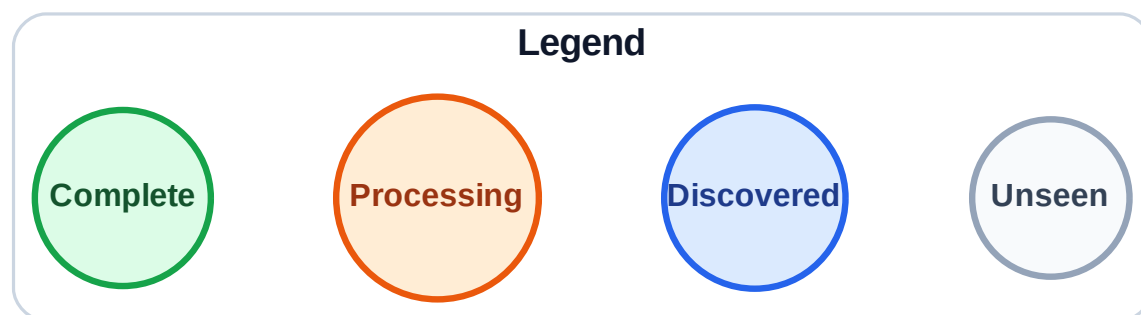
Stack (top at right): N | L | J | E | D | K | O | M | G | I | H

Visited: A, B, C, F



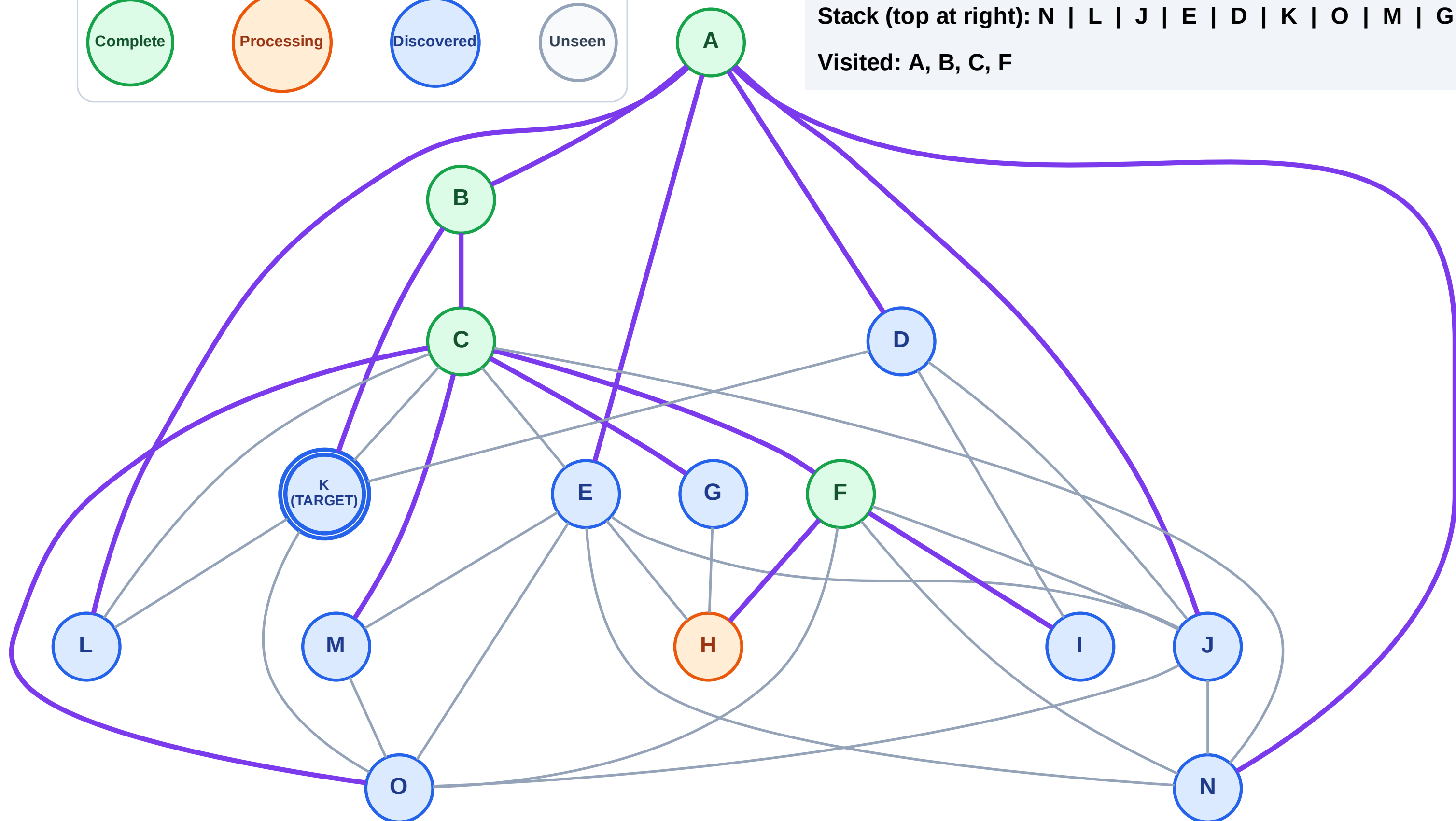
DFS Traversal

Step 10: Process node H



Stack (top at right): N | L | J | E | D | K | O | M | G | I

Visited: A, B, C, F



DFS Traversal

Step 11: No new neighbors from H

Legend

Complete

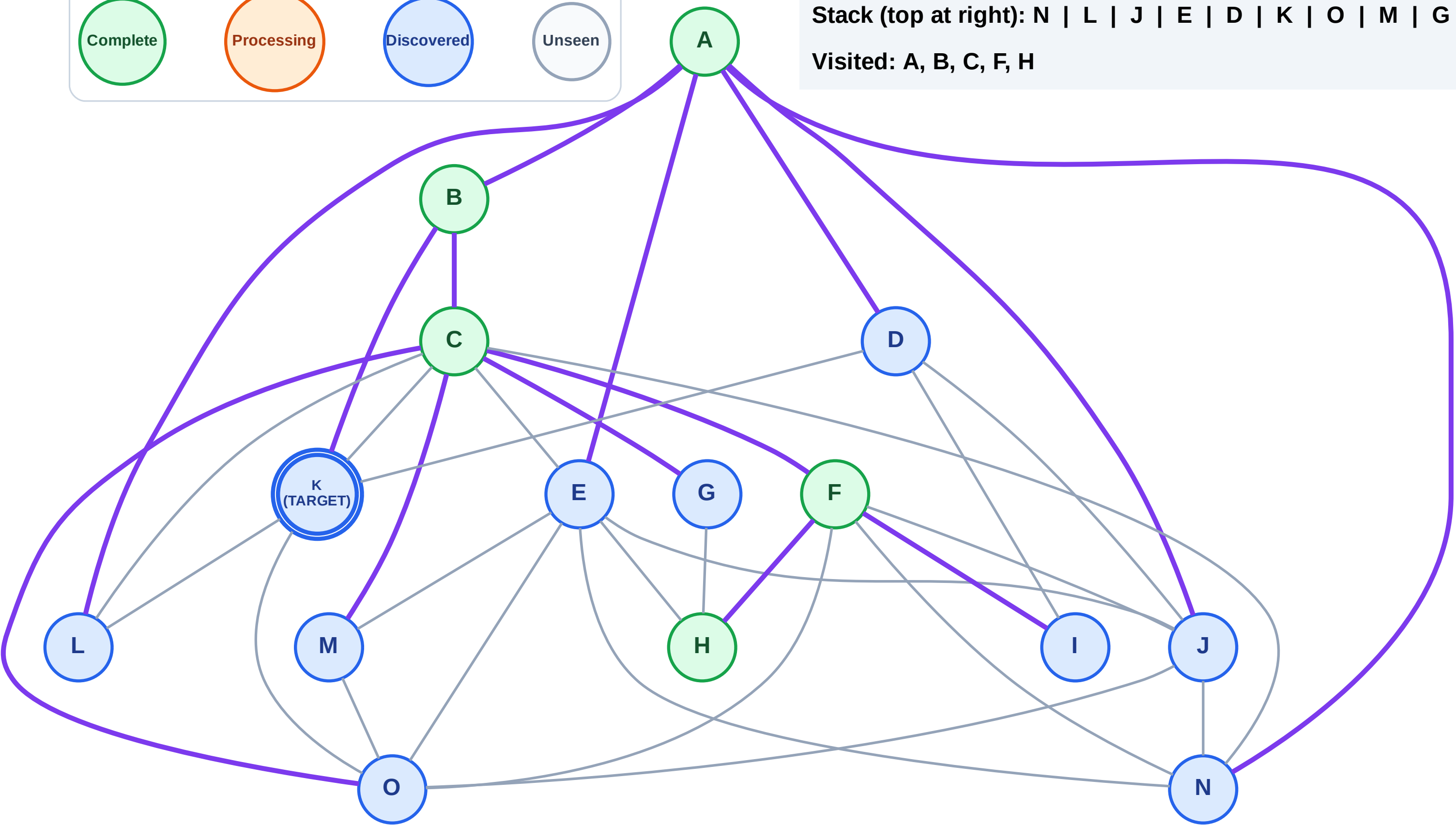
Processing

Discovered

Unseen

Stack (top at right): N | L | J | E | D | K | O | M | G | I

Visited: A, B, C, F, H



DFS Traversal

Step 12: Process node I

Legend

Complete

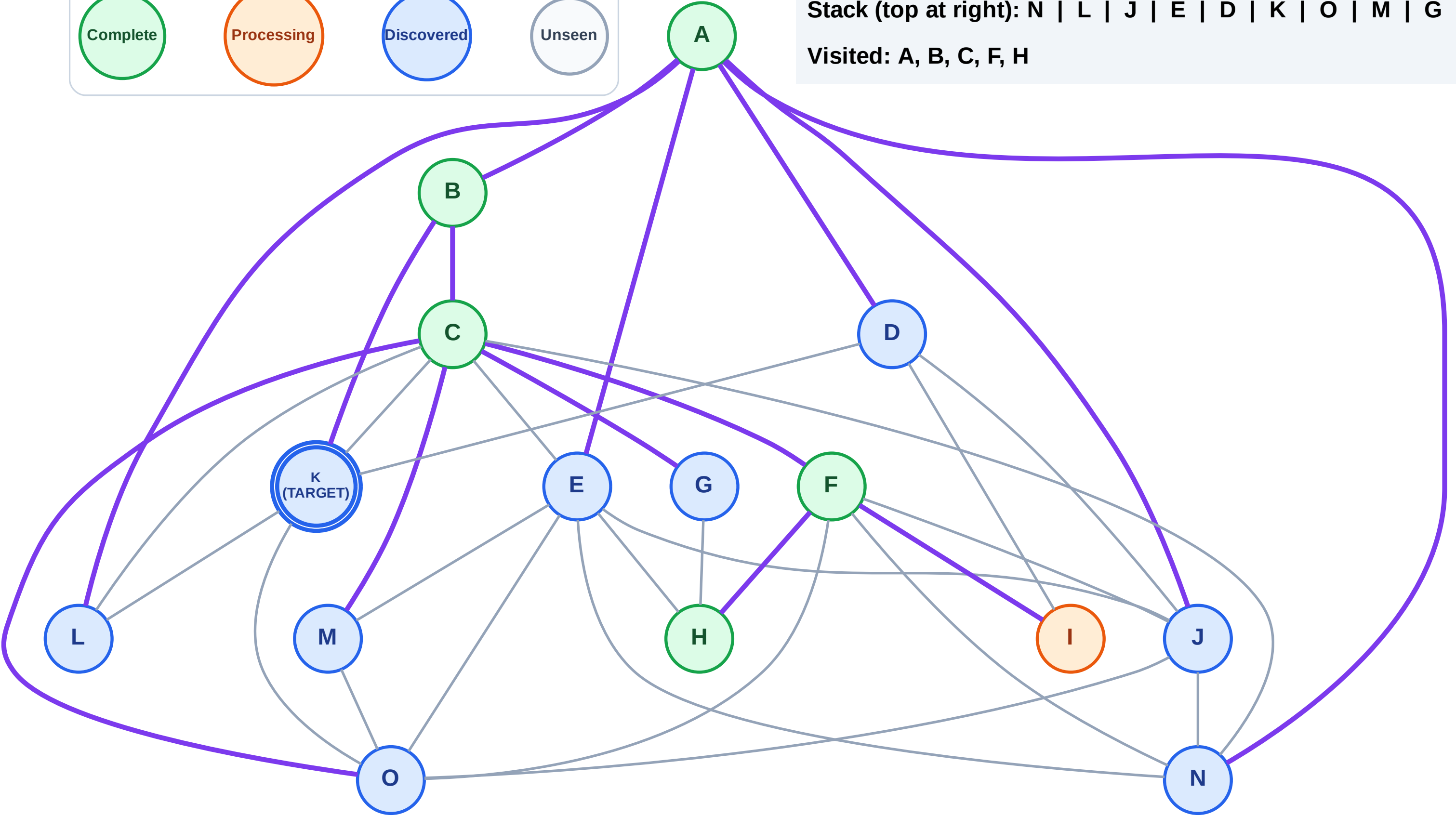
Processing

Discovered

Unseen

Stack (top at right): N | L | J | E | D | K | O | M | G

Visited: A, B, C, F, H



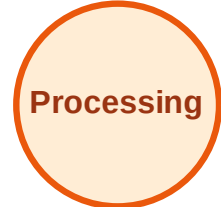
DFS Traversal

Step 13: No new neighbors from I

Legend



Complete



Processing



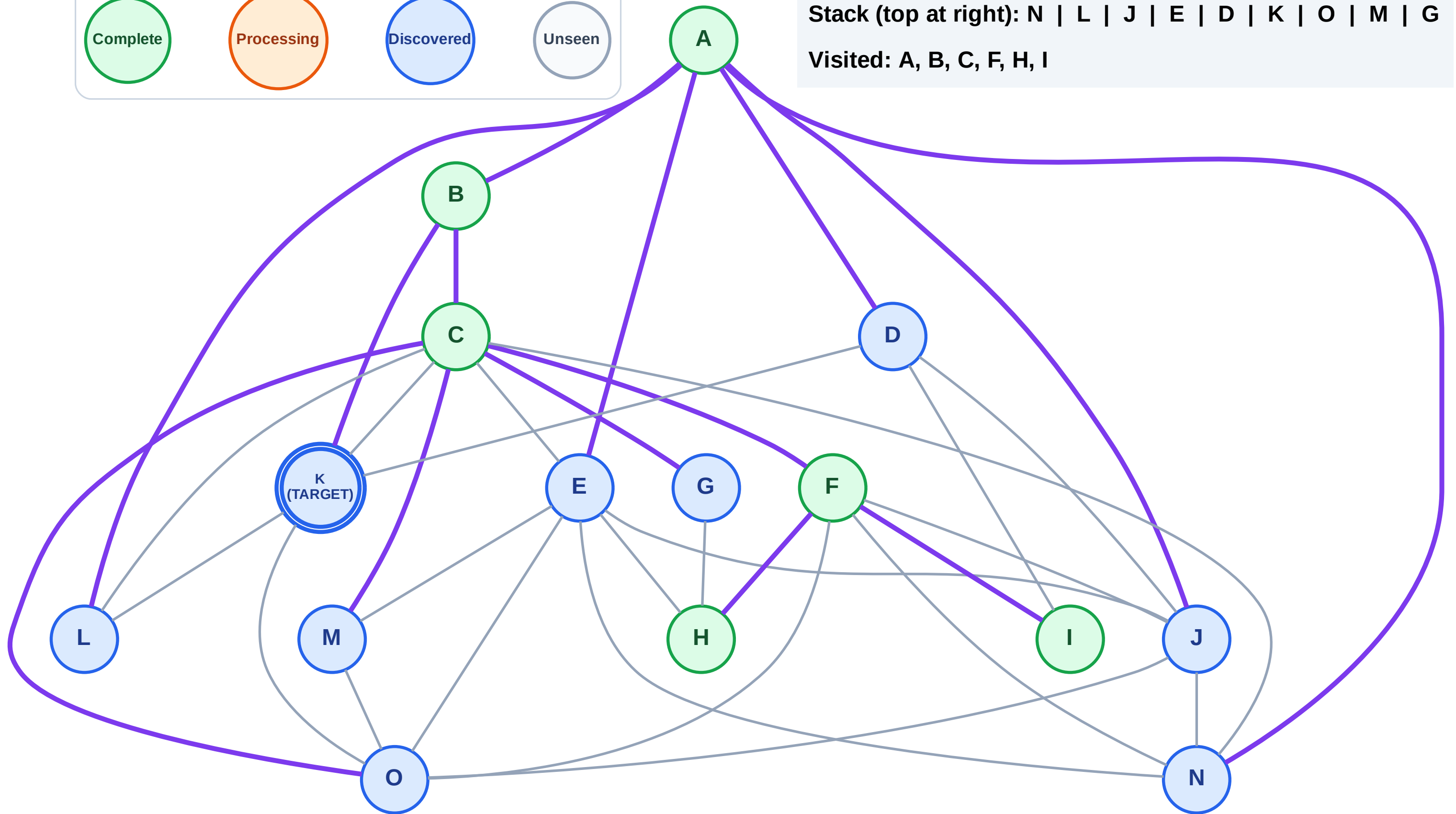
Discovered



Unseen

Stack (top at right): N | L | J | E | D | K | O | M | G

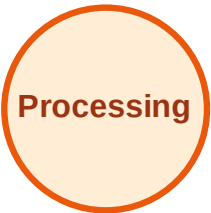
Visited: A, B, C, F, H, I



DFS Traversal

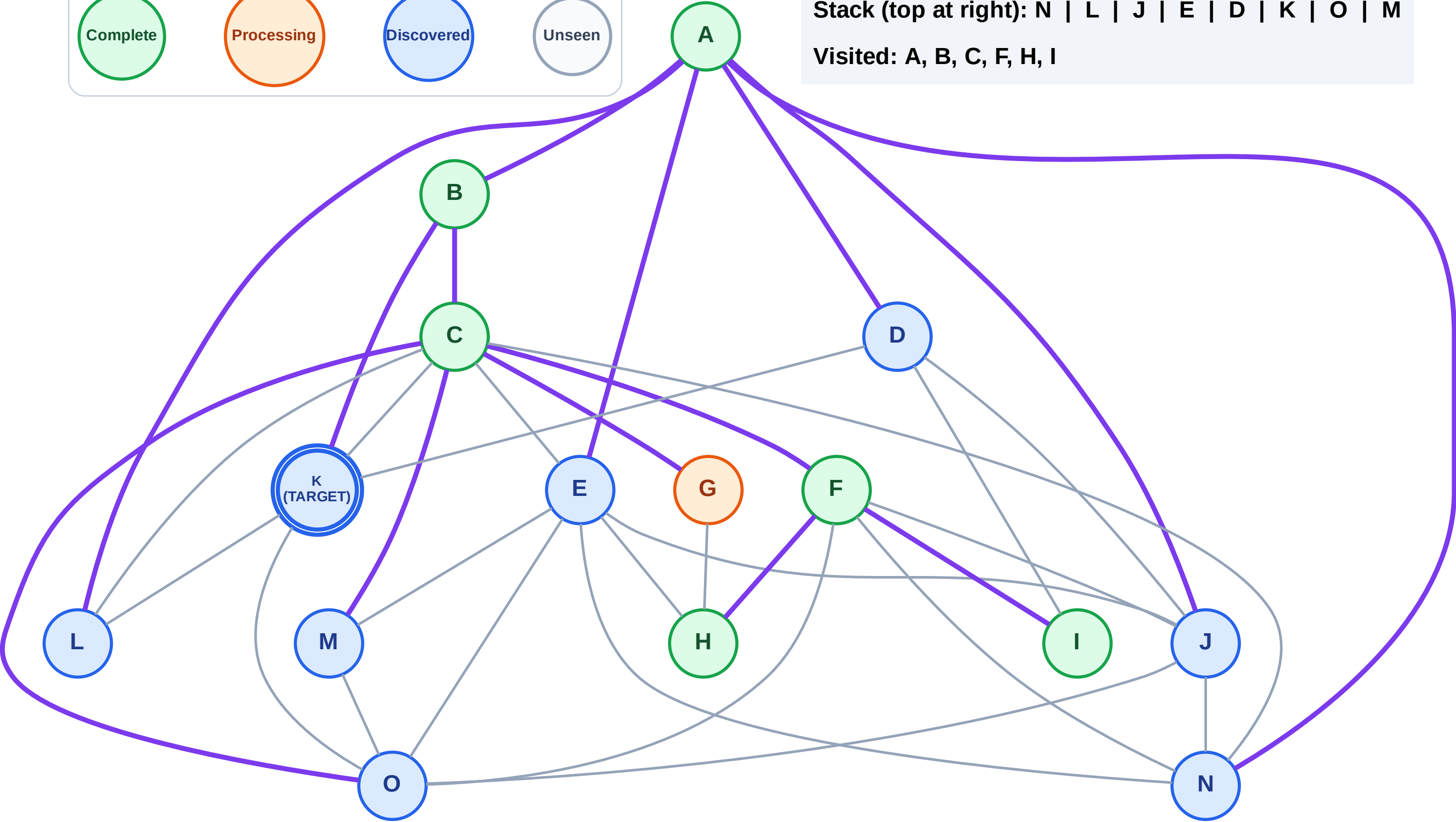
Step 14: Process node G

Legend



Stack (top at right): N | L | J | E | D | K | O | M

Visited: A, B, C, F, H, I



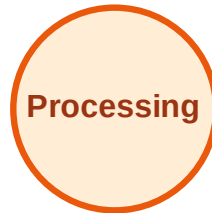
DFS Traversal

Step 15: No new neighbors from G

Legend



Complete



Processing



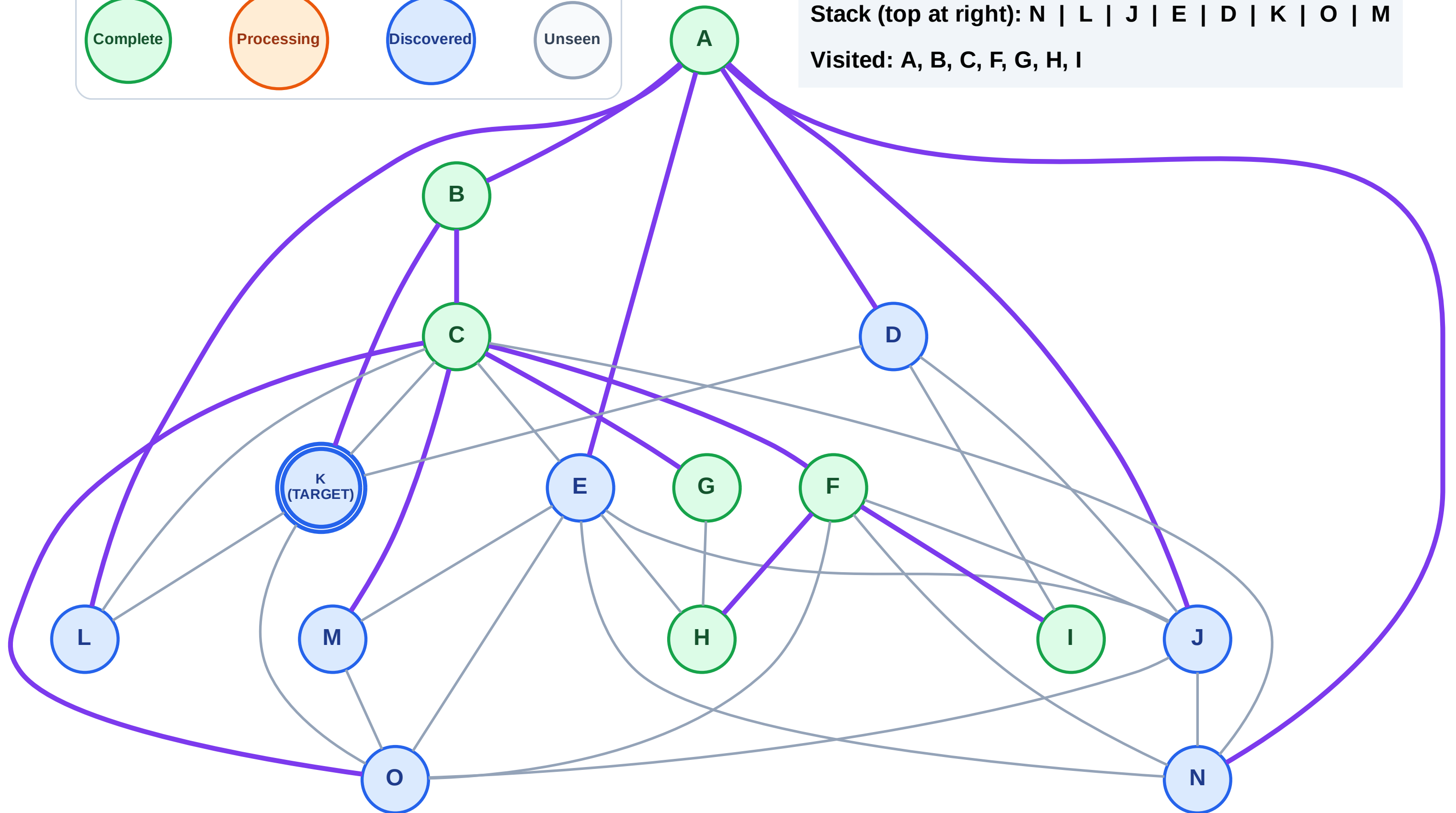
Discovered



Unseen

Stack (top at right): N | L | J | E | D | K | O | M

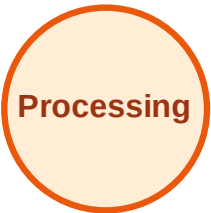
Visited: A, B, C, F, G, H, I



DFS Traversal

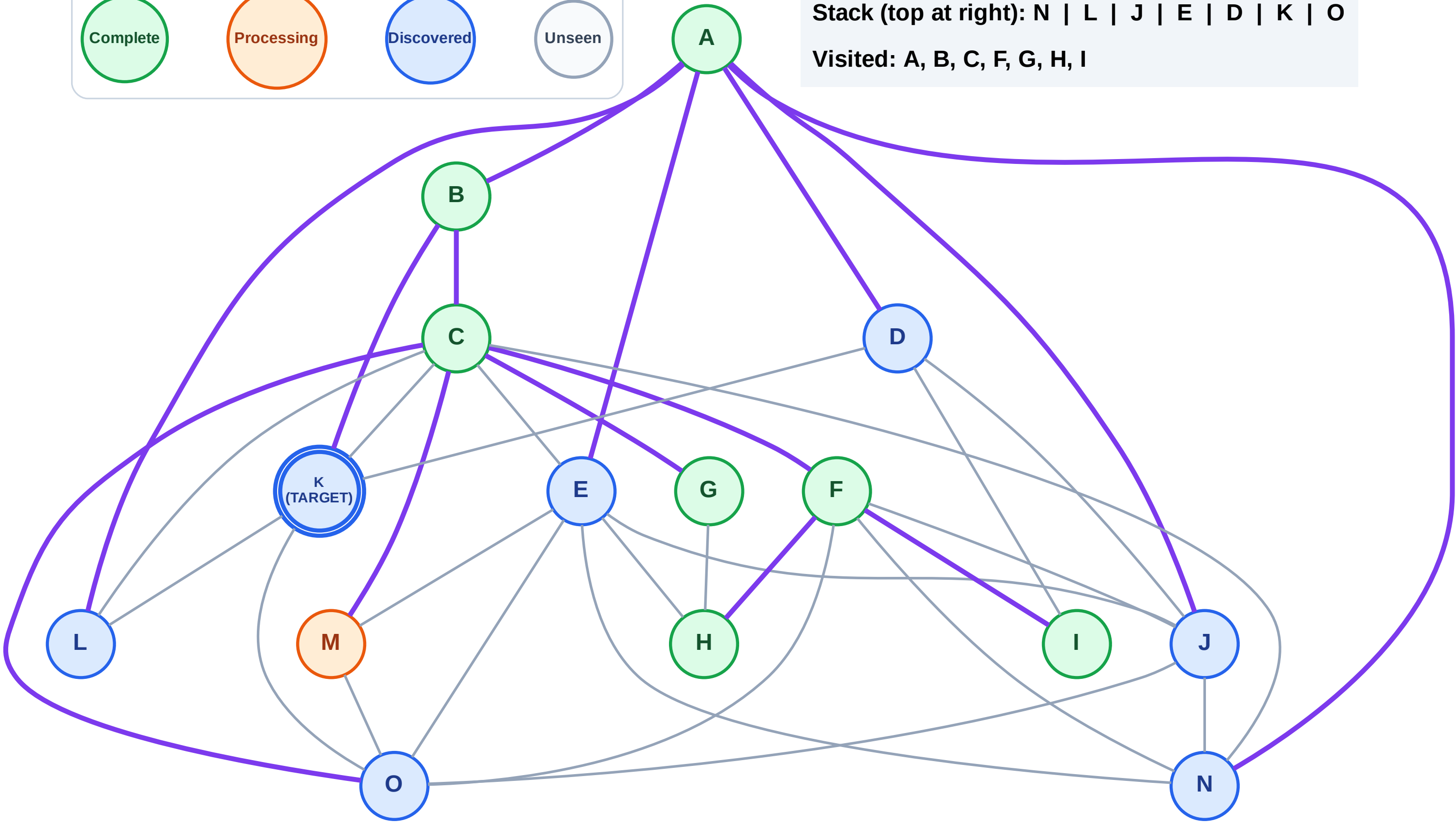
Step 16: Process node M

Legend



Stack (top at right): N | L | J | E | D | K | O

Visited: A, B, C, F, G, H, I



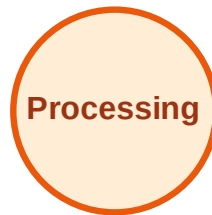
DFS Traversal

Step 17: No new neighbors from M

Legend



Complete



Processing



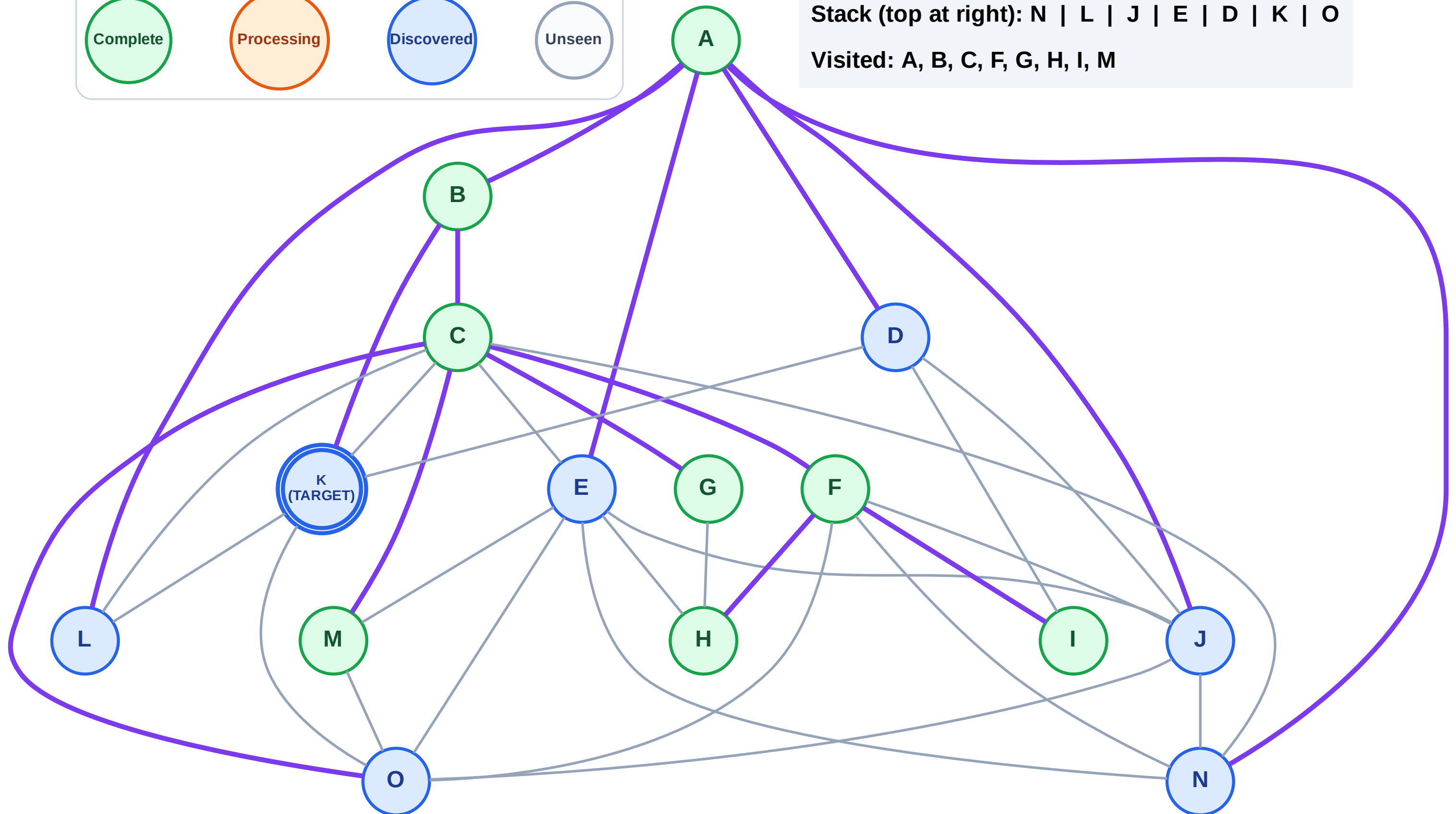
Discovered



Unseen

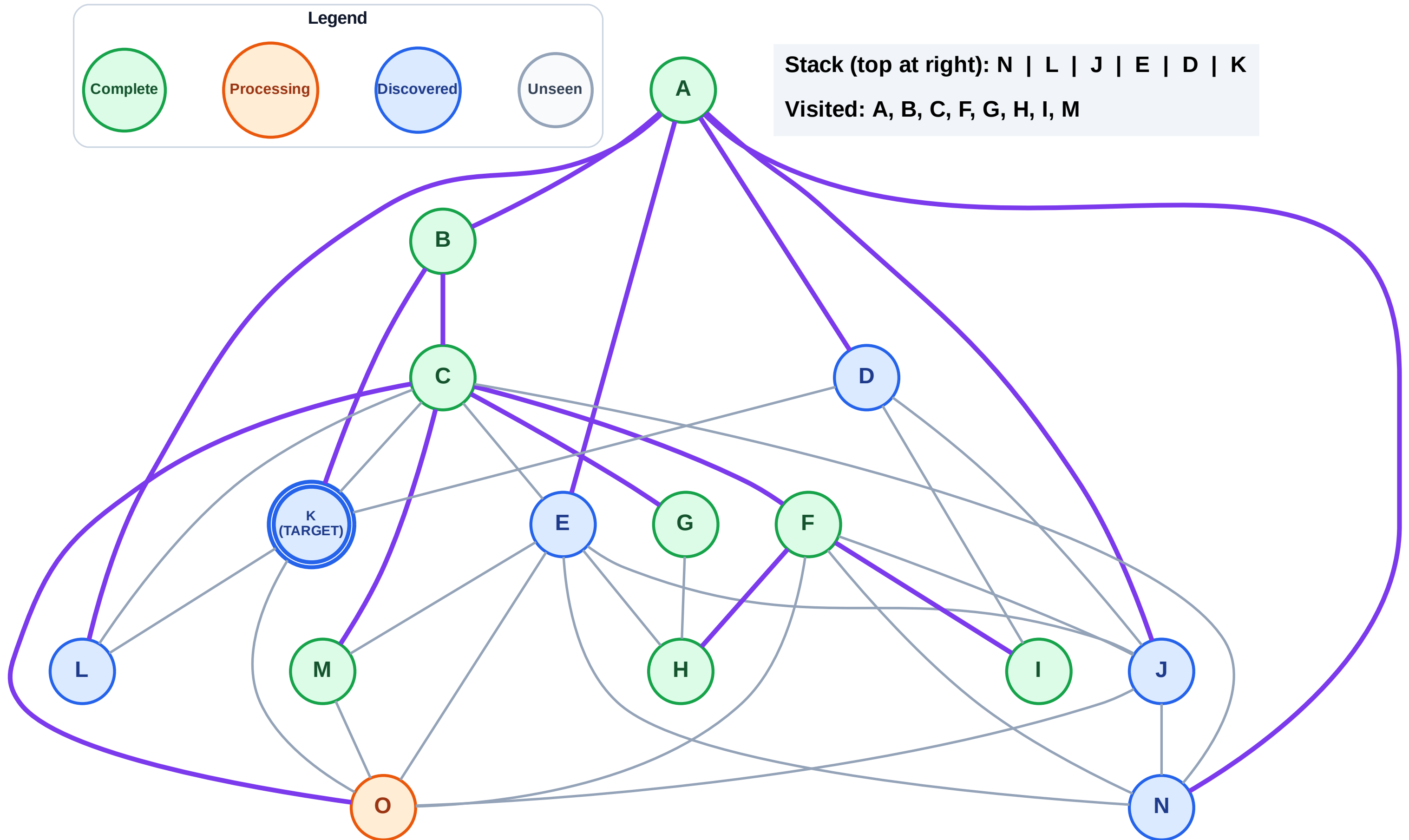
Stack (top at right): N | L | J | E | D | K | O

Visited: A, B, C, F, G, H, I, M



Step 18: Process node O

Step 18: Process node O



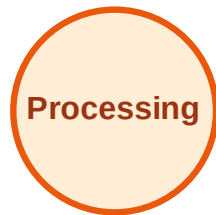
DFS Traversal

Step 19: No new neighbors from O

Legend



Complete



Processing



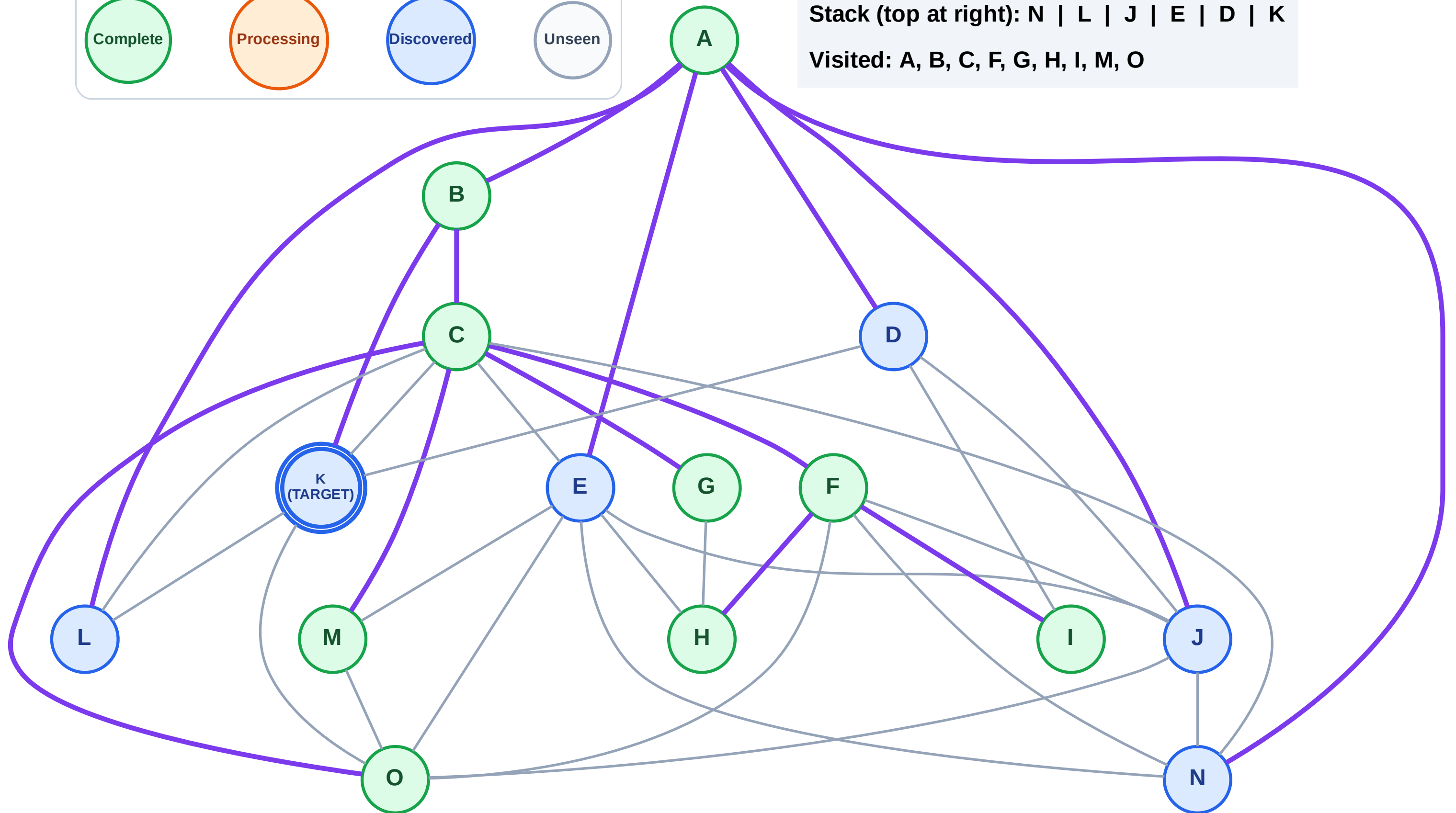
Discovered



Unseen

Stack (top at right): N | L | J | E | D | K

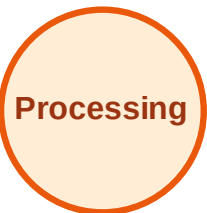
Visited: A, B, C, F, G, H, I, M, O



DFS Traversal

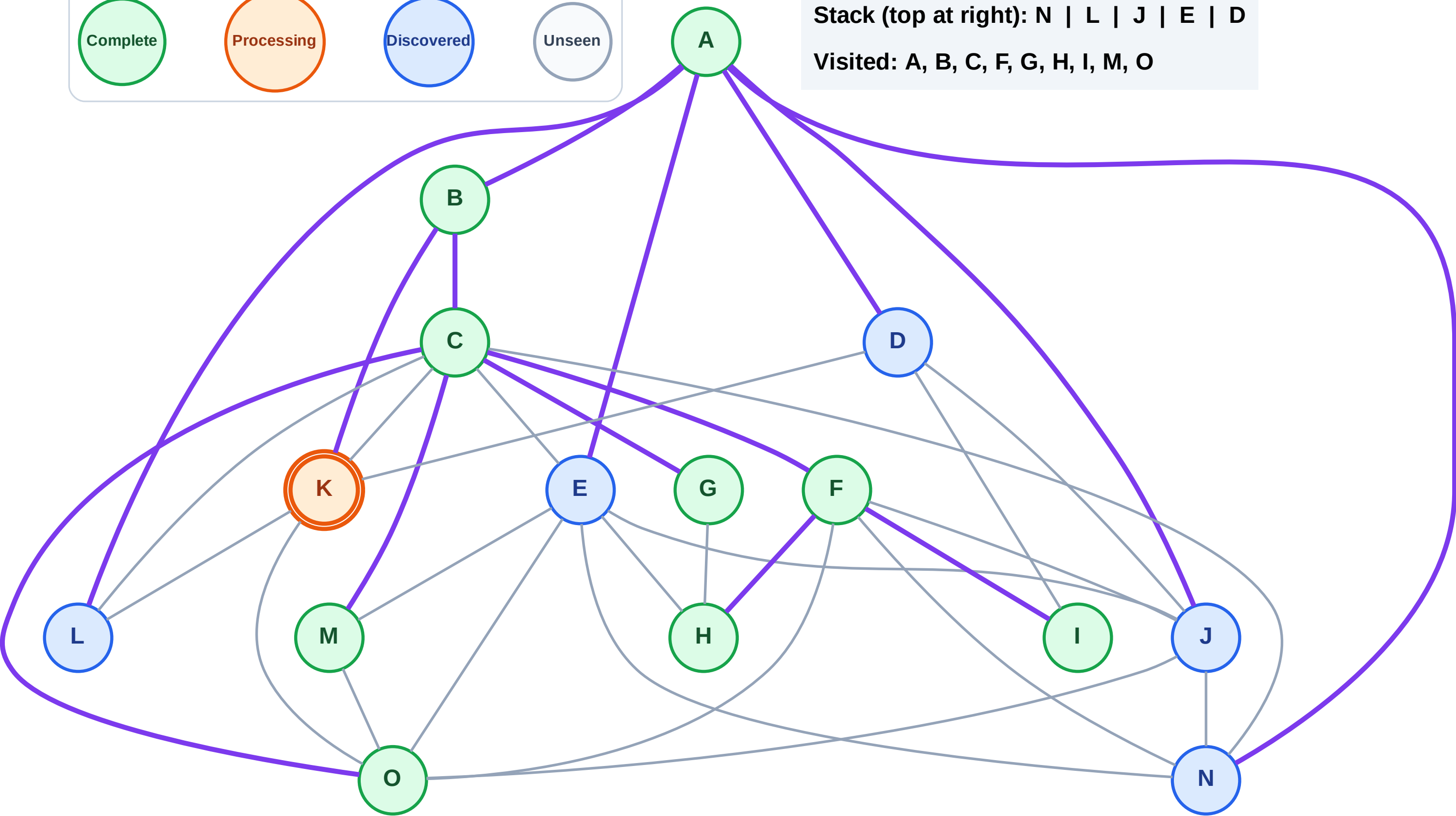
Step 20: Process node K

Legend



Stack (top at right): N | L | J | E | D

Visited: A, B, C, F, G, H, I, M, O



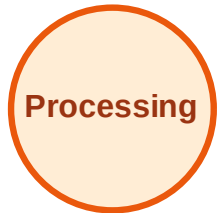
DFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): N | L | J | E | D

Visited: A, B, C, F, G, H, I, K, M, O

